

1 3 The cladding of a step-index optical fibre

[1 mark]

A reduces material dispersion.



B reduces signal loss.



C must have a higher refractive index than the core.



D must be opaque.



Monochromatic light of wavelength 610 nm illuminates a double slit. An interference pattern is observed on a screen 4.5 m from the slits. Consecutive bright fringes are 1.1 cm apart.

What is the slit separation?

[1 mark]

A 0.0025 mm



B 0.025 mm



C 0.25 mm



D 2.5 mm



Do not write

1 **7** A diffraction grating has a spacing of $3.0\ \mu\text{m}$

Light of wavelength $610\ \text{nm}$ is incident normally on the diffraction grating.

What is the largest order obtained?

[1 mark]

A 2nd

☐

B 4th

☐

C 5th

☐

D 8th

☐

Do not write

The refractive index for light passing from air into medium **A** is 1.3

The refractive index for light passing from air into medium **B** is 1.6

Which statement is **not** correct?

[1 mark]

A The critical angle for light travelling from **A** into **B** is about 54°

☐

B Light travelling from **B** into **A** will refract away from the normal.

☐

C The speed of light in **A** is about $0.77c$

☐

D Light travels slower in **B** than in **A**.

☐

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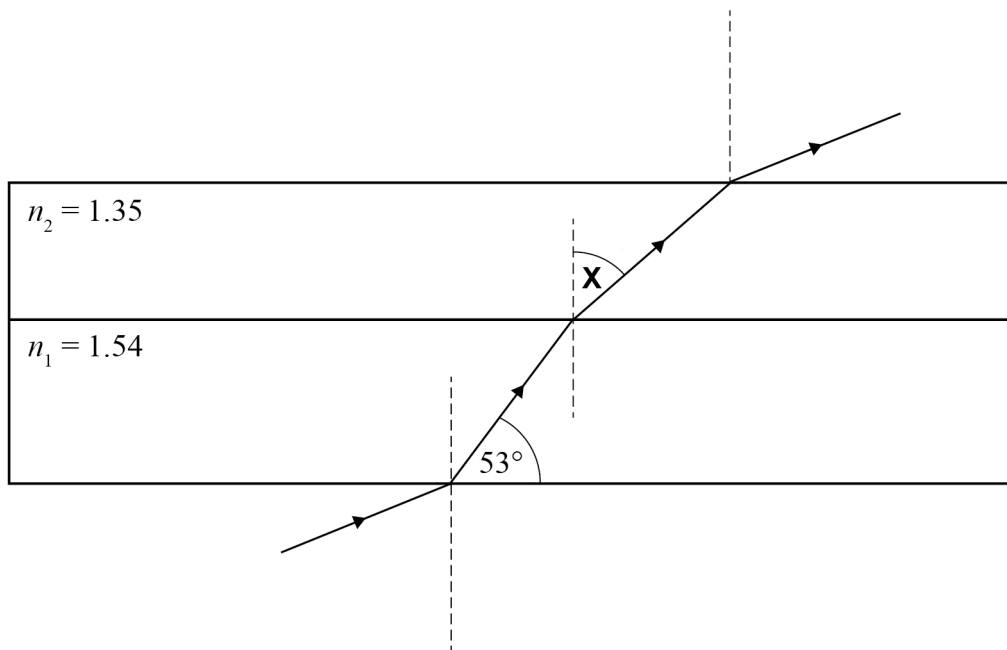


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The diagram shows the path of a ray of light through two transparent parallel layers.

The refractive indices shown on the diagram are for light travelling from air into each medium.

[1 mark]



What is angle X ?

A 32°

☐

B 37°

☐

C 43°

☐

D 67°

☐

Do not write

A student wishes to measure the slit separation of a double slit using the interference pattern produced by the slit.

Which row shows the source she should choose and the reason for her choice?

[1 mark]

	Source	Reason	
A	Monochromatic red light	The fringes will be brighter	☐
B	Monochromatic red light	The interference patterns do not overlap	☐
C	White light	The fringes will be brighter	☐
D	White light	The interference patterns do not overlap	☐

Turn over for the next question



Don't write

A filament lamp operates at a power of 21 W when there is a potential difference of 12 V across it.

Calculate the charge that flows through the filament lamp in 10 minutes.

[3 marks]

charge = _____ C



0 3 . 1 State **one** application of superconductors.

[1 mark]

0 3 . 2 Explain why superconductors are necessary in the application given in question **03.1**.

[1 mark]

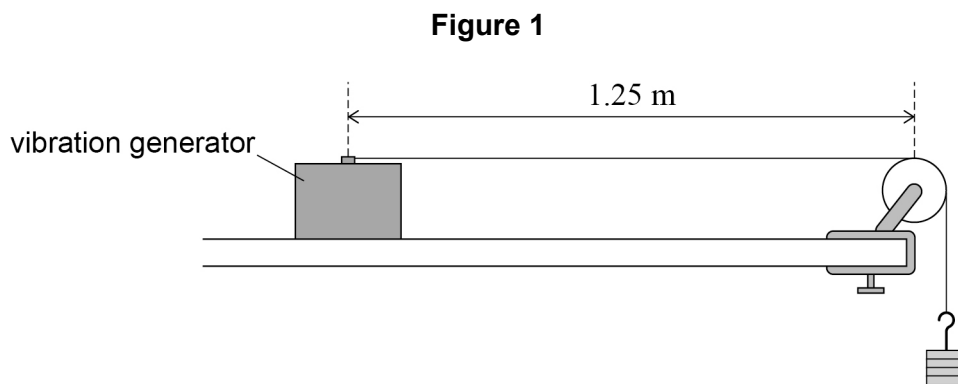
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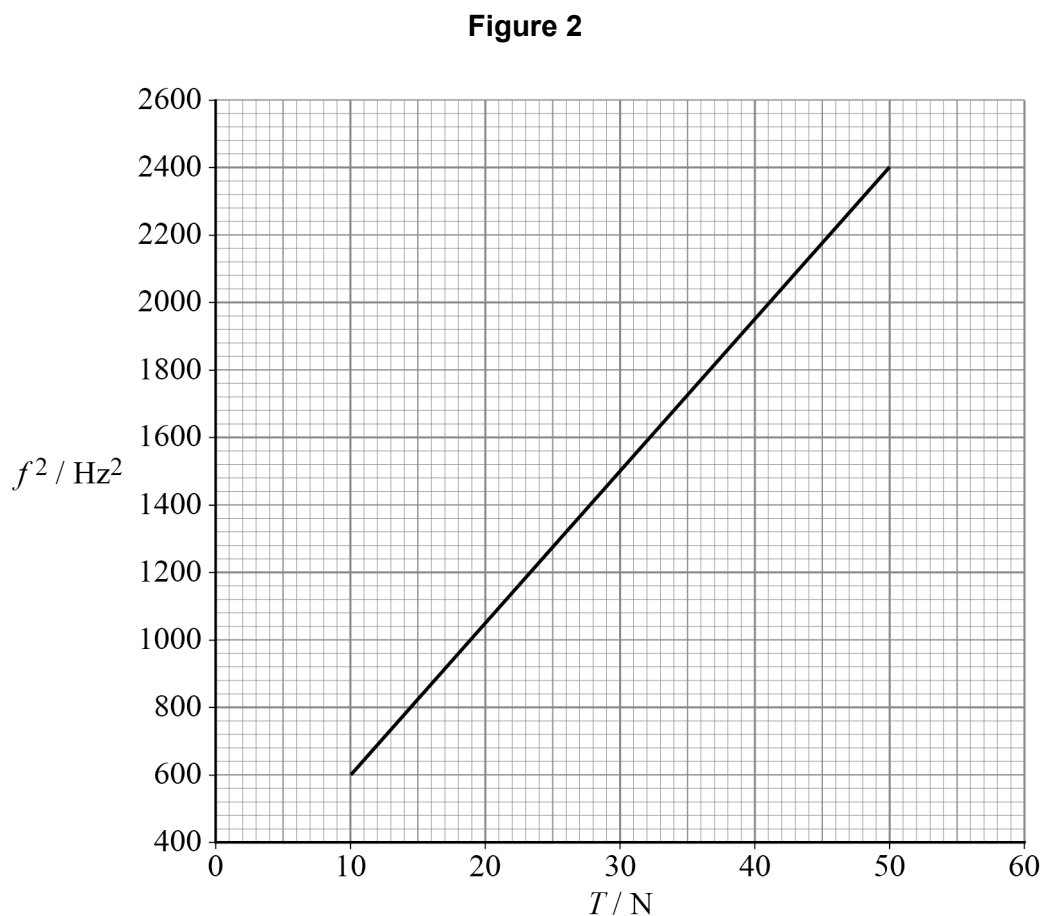
Figure 1 shows an experimental arrangement used to investigate stationary waves on a stretched string.



The string was made to vibrate. The vibrating length l of the string was kept constant at 1.25 m.

The frequency f of the first harmonic of the vibrating string was obtained as the tension T in the string was increased.

Figure 2 shows the variation of f^2 with T .



0 4 . 1

Show that the mass per unit length, μ , of the string is equal to $\frac{1}{4l^2 \text{gradient}}$

[2 marks]

0 4 . 2

Determine the mass of 1.25 m of the string.

[4 marks]

mass = _____ kg

6

0 5

Describe how ultraviolet radiation is produced inside a fluorescent tube.

[3 marks]

3



Figure 3 shows the variation of resistance with temperature for a negative temperature coefficient (ntc) thermistor.

Figure 3

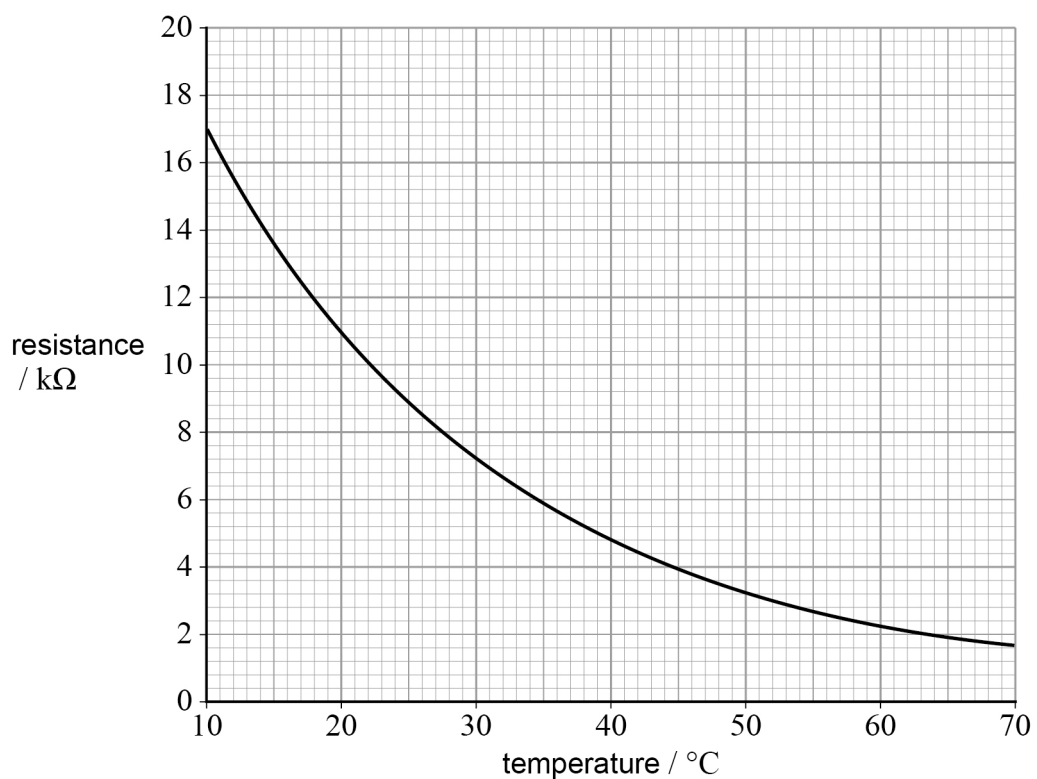
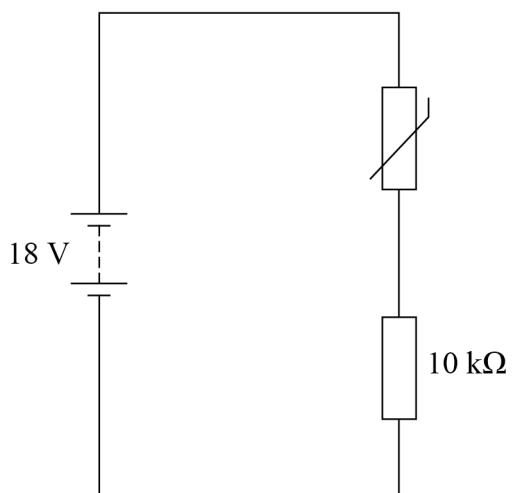


Figure 4 shows the ntc thermistor connected in series with a $10\text{ k}\Omega$ resistor and a battery of emf 18 V . The temperature of the thermistor is $50\text{ }^{\circ}\text{C}$. The battery has negligible internal resistance.

Figure 4



0 6 . 1 Determine the current in the circuit.

State an appropriate unit for your answer.

[4 marks]

current = _____

unit = _____

0 6 . 2 Calculate the rate of energy transfer by the 10 k Ω resistor.

[2 marks]

rate of energy transfer = _____ W



0 7 . 1 Define the electron volt.

[1 mark]

0 7 . 2 An electron, in a vacuum, is accelerated from rest through a potential difference V .

Show that the final speed v of the electron is given by

$$v = 5.9 \times 10^5 \sqrt{V}$$

Assume the mass of the electron is constant.

[2 marks]

0 7 . 3 An electron is accelerated from rest through a potential difference $V = 5.0$ kV.

Calculate the minimum de Broglie wavelength for the electron.

[2 marks]

minimum wavelength = _____ m



0 7 . 4

Accelerated electrons can be used to produce diffraction patterns. Electron diffraction was first observed in 1927.

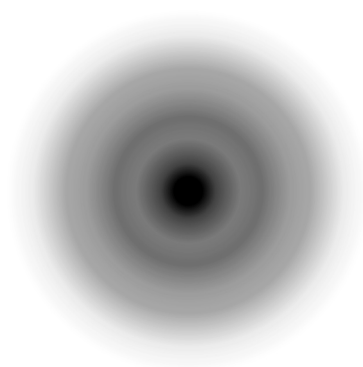
Discuss why this observation led to a development in the scientific ideas about the nature of the electron.

[2 marks]

0 7 . 5

Figure 5 shows the pattern of rings produced in an electron diffraction tube by electrons that have been accelerated by a potential difference.

Figure 5



Explain why the diameter of these rings increases when the potential difference is decreased.

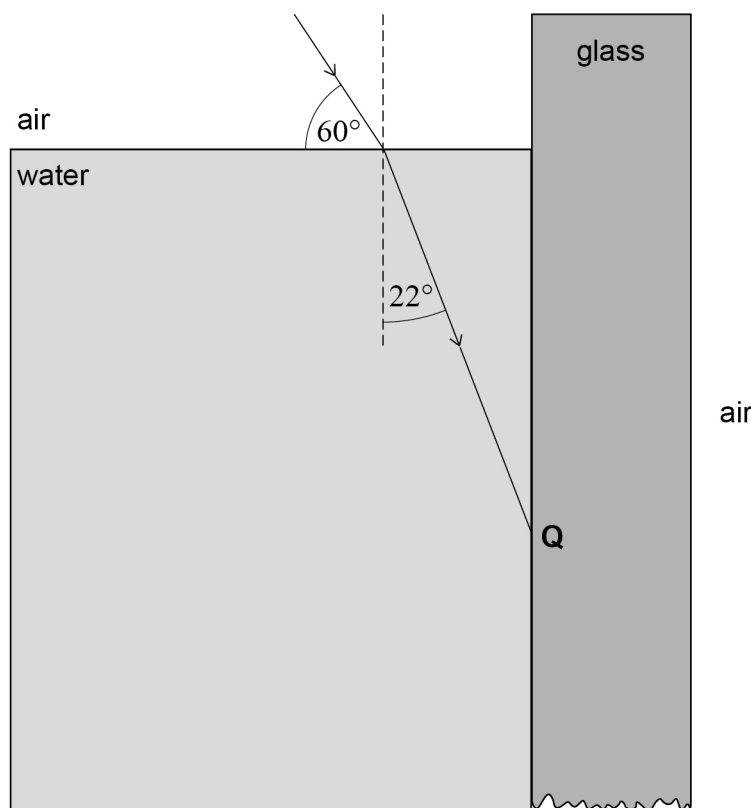
[2 marks]

08

A teacher shines a laser beam onto the surface of water in a glass container. **Figure 6** shows a ray entering the water. The angle between the ray and the water surface is 60° . The refractive index of the glass is 1.55

Figure 6

not to scale



08

1

Show that the refractive index of the water is approximately 1.3

[2 marks]

08

2

The ray refracts at the vertical water–glass boundary at **Q** on **Figure 6**.

Explain why the ray cannot undergo total internal reflection at **Q**.

[1 mark]



0 8 . 3 Determine the angle of refraction of the ray at **Q**.

[3 marks]

angle of refraction = _____ degrees

0 8 . 4 The ray passes from **Q** to the glass–air boundary.

Deduce what happens to the ray immediately after striking the glass–air boundary.

[3 marks]

0 8 . 5 A class of students observe the demonstration.

Suggest **one** safety precaution the teacher should take when using the laser.

[1 mark]



Figure 7 shows three simple pendulums suspended from a ceiling.

Table 1 shows the properties of each pendulum.

Figure 7

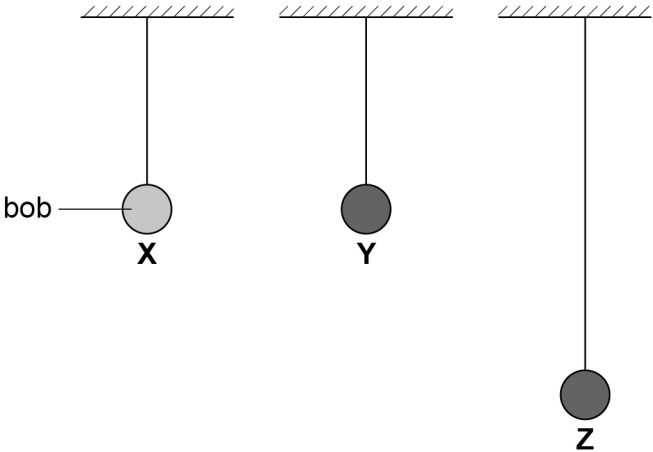


Table 1

Pendulum	Length of pendulum / cm	Mass of pendulum bob / g
X	150	100
Y	150	200
Z	300	200



0 9 . 1

The three pendulum bobs are all pulled forward by the same small horizontal distance and then released from rest at the same time.
Ignore air resistance in this part of the question.

Compare the oscillations of the three pendulums.

[3 marks]

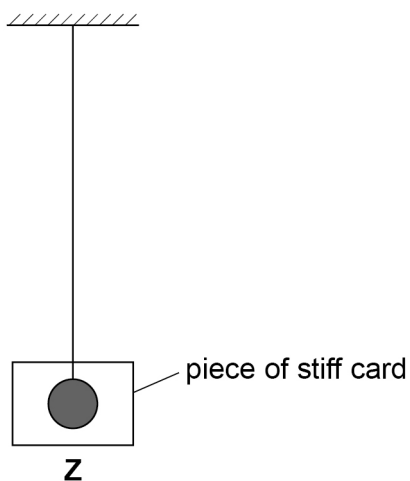
Question 9 continues on the next page



0 9 . 2

A piece of stiff card, of negligible mass, is attached to pendulum **Z** as shown in **Figure 8**. The card damps the motion of the pendulum.

Figure 8



The pendulum bob is pulled forward by the same small horizontal distance as in question **09.1** and released from rest.

Explain the effect of damping on the oscillations of pendulum **Z**.

[2 marks]



Gravitational field strength g varies between different locations on the Earth's surface. **Table 2** shows the value of g for two locations.

Table 2

Location	$g / \text{N kg}^{-1}$
Iceland	9.85493
Kenya	9.77385

0 9 . 3

A simple pendulum of length 0.450 m is set into oscillation in Iceland.

Calculate the frequency of oscillation of the pendulum.

[2 marks]

frequency = _____ Hz

0 9 . 4

Explain how a simple pendulum in Kenya could be made to oscillate at the same frequency as the simple pendulum in Iceland.

You may wish to support your answer with a calculation.

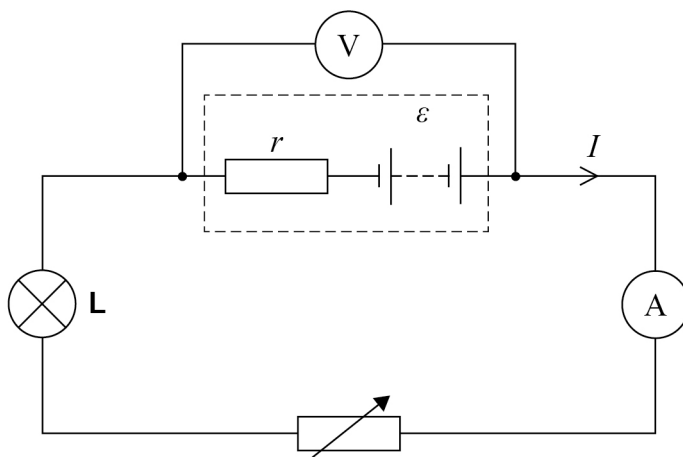
[2 marks]



Section B

Answer **all** questions in this section.

1 0

A battery has an emf of ε and an internal resistance of r .**Figure 9** shows a circuit used to investigate the relationship between the terminal pd V of the battery and the current I in the external circuit.**Figure 9**

1 0 . 1

Suggest why the lamp **L** was included in the circuit.**[1 mark]**

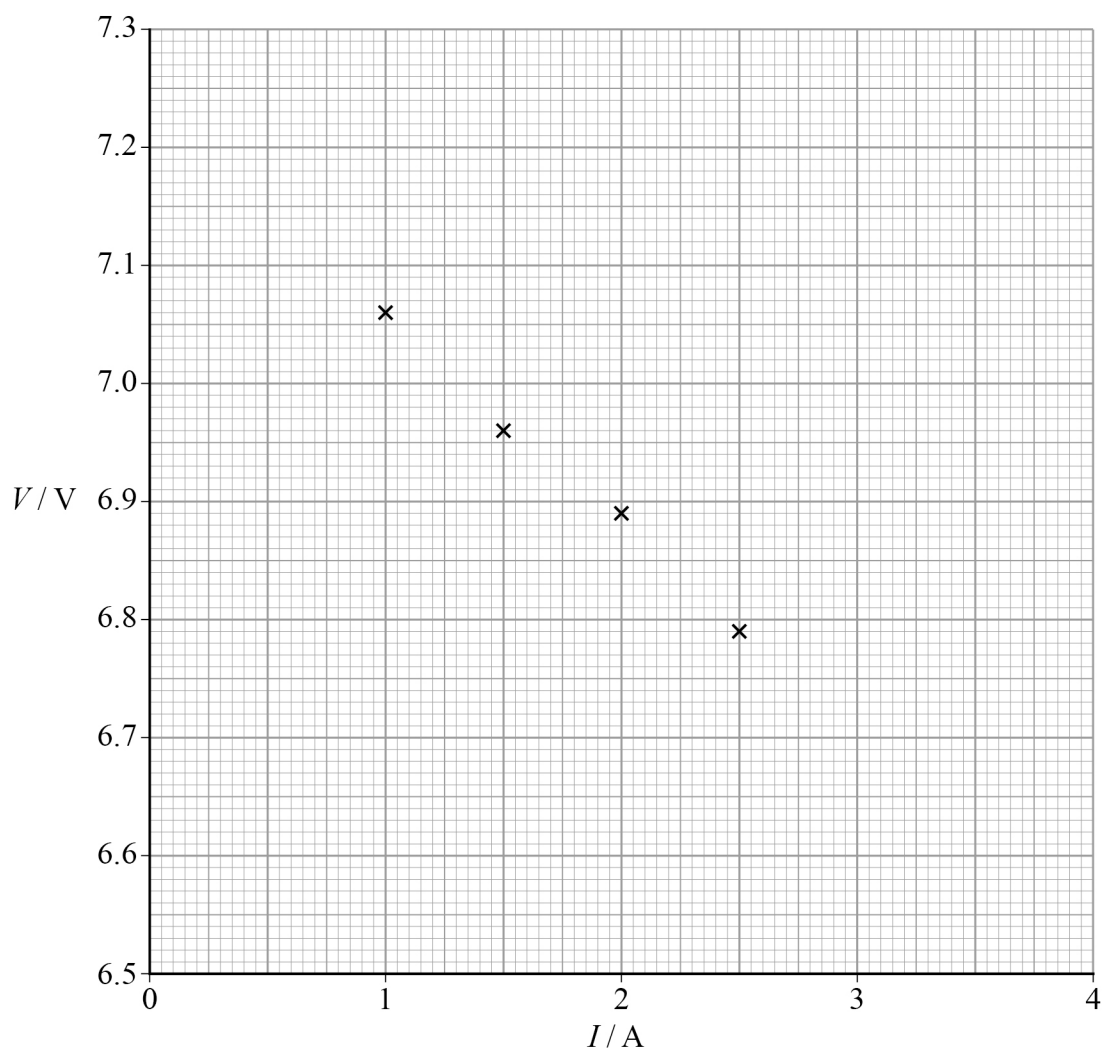
1 0 . 2

Table 3 shows results from this experiment. The uncertainty in V was ± 0.02 V. The uncertainty in I was negligible.**Table 3**

I / A	1.0	1.5	2.0	2.5	3.0	3.5
V / V	7.06	6.96	6.89	6.79	6.72	6.62

Some of these data points are plotted on **Figure 10**.Plot the remaining **two** points and add error bars to **all** the points.**[2 marks]**

Figure 10



1 0 . 3 Draw a line of best fit on **Figure 10**.

[1 mark]

1 0 . 4 Determine the gradient of your line of best fit.

[2 marks]

gradient = _____ V A^{-1}

Question 10 continues on the next page

Turn over ►



10.5 The equation that relates V and I is

$$V = \varepsilon - I r$$

Determine the values of ε and r .

[2 marks]

$$\varepsilon = \text{_____ V}$$

$$r = \text{_____ } \Omega$$

8



Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

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Figure 11 shows how an ultrasound scanner can be used to determine the depth of internal organs below the skin. A transducer emits an ultrasound pulse and then detects reflections of the pulse from internal organs. The reflected pulses are displayed on an oscilloscope screen.

Figure 11

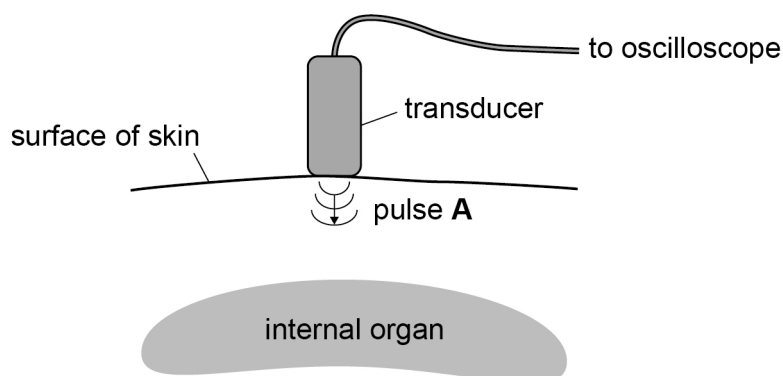
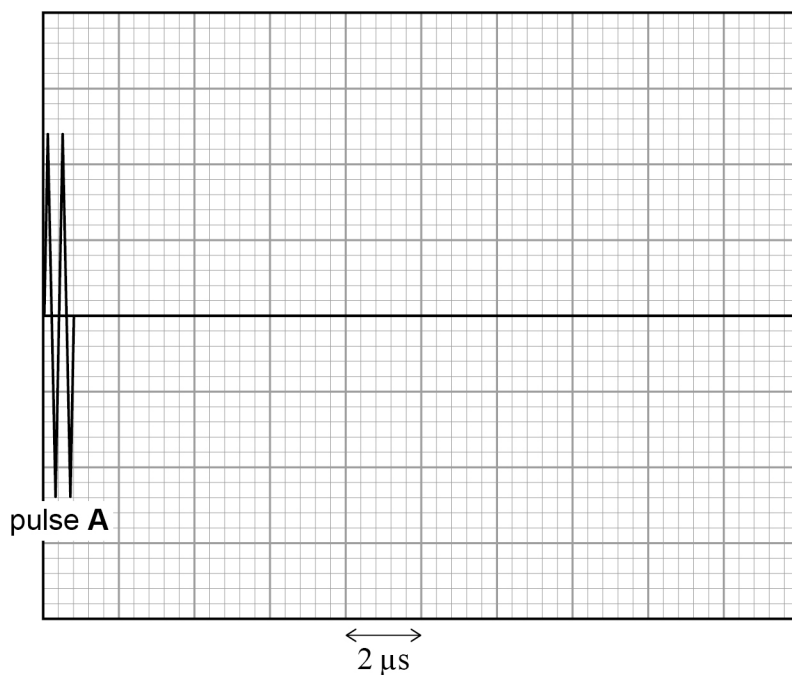


Figure 12 shows the oscilloscope display for an ultrasound pulse **A**. The time-base of the oscilloscope is set to $2\ \mu\text{s}$ per division.

Figure 12



1	1
---	---

.

1

 Determine the frequency of the ultrasound waves in pulse **A**.

[3 marks]

frequency = _____ Hz

1	1
---	---

.

2

 One reflection of pulse **A** is detected $12.6\ \mu\text{s}$ after it was transmitted.

Sketch on **Figure 12** the appearance of the reflection of pulse **A**.

[2 marks]

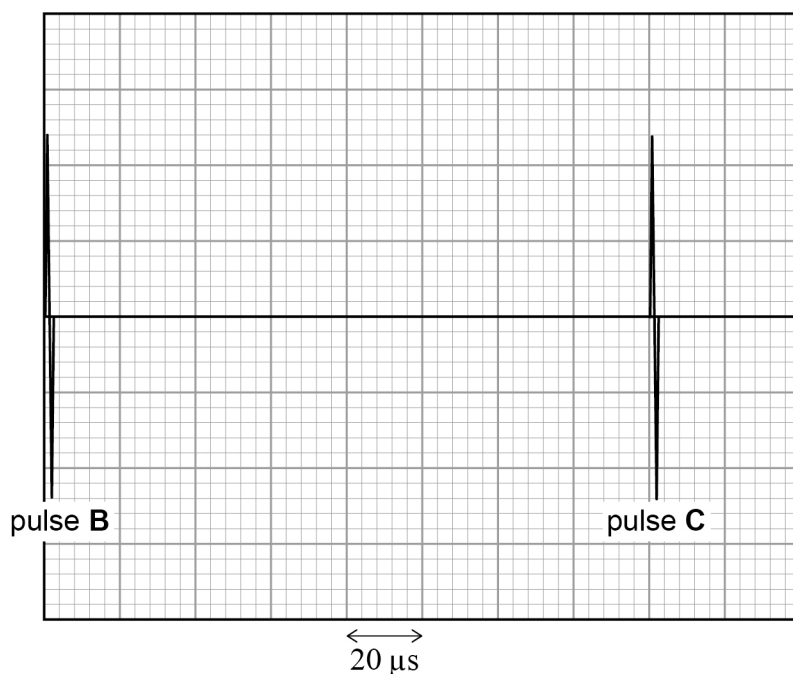
Question 11 continues on the next page



1 1 . 3 The scanner emits ultrasound pulses at regular time intervals.

Figure 13 shows an oscilloscope display of two successive pulses, **B** and **C**, emitted from the scanner. The time-base of the oscilloscope is now set to $20\ \mu\text{s}$ per division.

Figure 13



The reflections of pulse **B** must be detected before pulse **C** is emitted, so there is a maximum depth that can be measured for an internal organ.

Calculate this maximum depth.

$$\text{speed of ultrasound in body tissue} = 1540\ \text{m s}^{-1}$$

[3 marks]

maximum depth = _____ m

Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each answer completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown. 

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown. 

You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional sheets for this working.

1 2 The table shows the values of V and corresponding values of I for components **A**, **B**, **C** and **D**.

	A	B	C	D
V / V	I / A	I / A	I / A	I / A
0	0.0	0.0	0.0	0.0
2	0.4	0.9	0.2	0.0
4	0.8	1.5	0.4	0.1
6	1.2	1.9	0.6	0.7
8	1.6	2.1	0.8	1.4
10	2.0	2.2	1.0	2.1

Which component is an ohmic conductor with the lowest resistance?

[1 mark]

A ☐

B ☐

C ☐

D ☐

Turn over ►



- 1 3** A metal wire has resistance R . The wire is cut in half and the two cut pieces are joined in parallel to form one component.

What is the resistance of the component?

[1 mark]

A $4R$ ☐

B $2R$ ☐

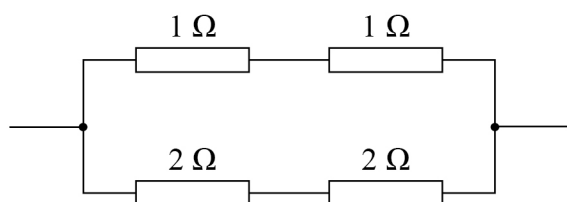
C $\frac{R}{2}$ ☐

D $\frac{R}{4}$ ☐

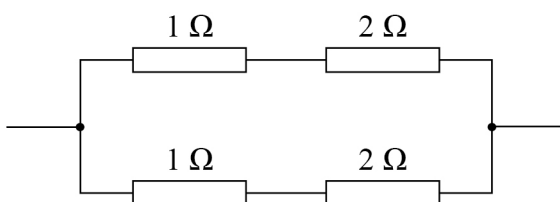
- 1 4** Which network of resistors has the largest total resistance?

[1 mark]

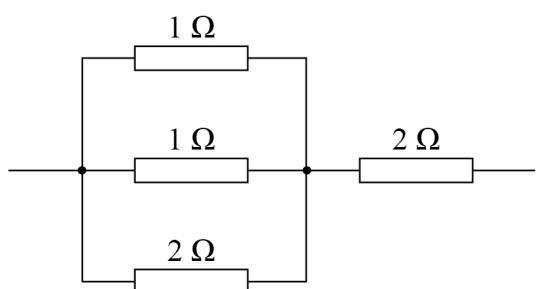
A



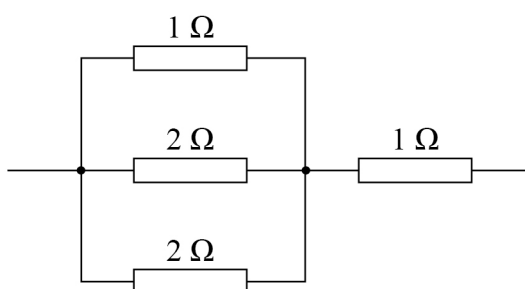
B



C



D



A ☐

B ☐

C ☐

D ☐



- 1 5** A solid copper cylinder has volume $1.3 \times 10^{-4} \text{ m}^3$ and length 15 cm. Copper has a resistivity of $1.7 \times 10^{-8} \Omega \text{ m}$.

What is the resistance between the two ends of the copper cylinder?

[1 mark]

A $2.9 \times 10^{-6} \Omega$

☐

B $2.0 \times 10^{-5} \Omega$

☐

C $2.0 \times 10^{-3} \Omega$

☐

D $2.9 \times 10^{-2} \Omega$

☐

- 1 6** What is a unit for potential difference?

[1 mark]

A $\text{A } \Omega^{-1}$

☐

B C J^{-1}

☐

C $\text{J A}^{-1} \text{ s}^{-1}$

☐

D W A

☐

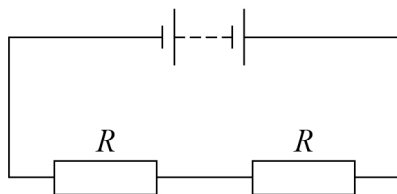
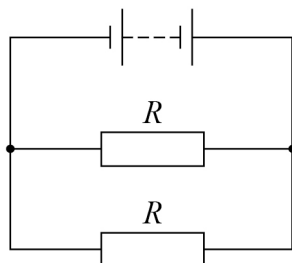
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1 7

The components in circuits **X** and **Y** are identical. The cells have negligible internal resistance.

circuit **X**circuit **Y**

What is the ratio $\frac{\text{power dissipated in circuit X}}{\text{power dissipated in circuit Y}}$?

[1 mark]**A** 0.25
☐
B 0.50
☐
C 2.00
☐
D 4.00
☐
1 8

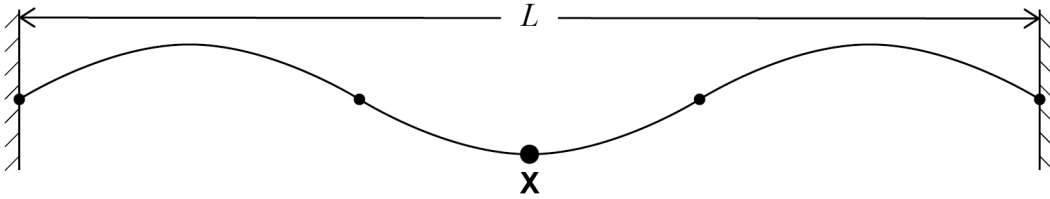
A mass oscillates on a spring **P** with a time period of 2.00 s. Spring **P** has stiffness k . The same mass now oscillates on spring **Q** of stiffness $3k$.

What is the time period of the oscillation of the mass on spring **Q**?

[1 mark]**A** 0.67 s
☐
B 1.15 s
☐
C 1.73 s
☐
D 3.46 s
☐


1 9Do not write
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box

A string oscillates at the third harmonic within a cavity of length L .
The speed of the wave is c .



Point **X** is at an antinode.

What is the shortest time before **X** returns to the same position?

[1 mark]

A $\frac{2L}{3c}$

☐

B $\frac{3L}{2c}$

☐

C $\frac{2c}{3L}$

☐

D $\frac{3c}{2L}$

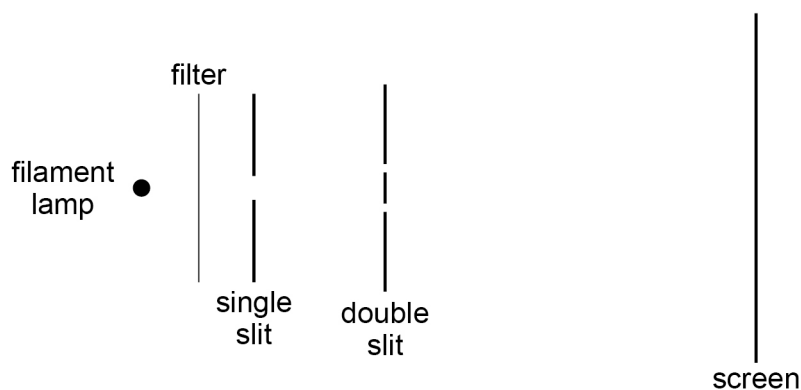
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2 0 The diagram shows a demonstration of Young's double slit experiment.



Which change would increase the fringe separation observed on the screen?

[1 mark]

A Moving the lamp to the right

☐

B Moving the single slit to the left

☐

C Moving the double slit to the right

☐

D Moving the screen to the right

☐

2 1 Monochromatic light passes through a diffraction grating that has a line spacing of 2×10^{-6} m. The angle between the two $n = 2$ maxima is 64° .

What is the wavelength of this light?

[1 mark]

A 5.3×10^{-7} m

☐

B 9.0×10^{-7} m

☐

C 1.1×10^{-6} m

☐

D 1.8×10^{-6} m

☐

2 2 What change in the design of an optical fibre would reduce modal dispersion?

[1 mark]

A Decreasing the refractive index of the cladding

☐

B Increasing the refractive index of the core

☐

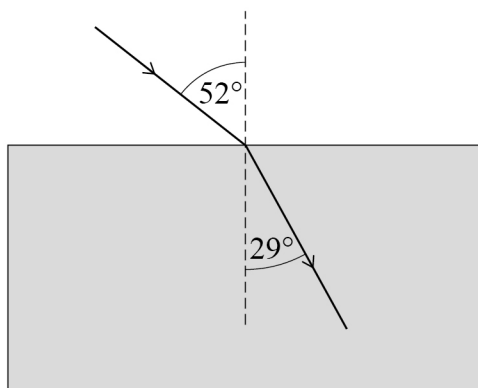
C Making the core narrower

☐

D Making the core wider

☐


2 3 A ray of light passes from air into a transparent material.



What is the speed of light in the transparent material?

[1 mark]

A $8.8 \times 10^6 \text{ m s}^{-1}$

☐

B $1.7 \times 10^8 \text{ m s}^{-1}$

☐

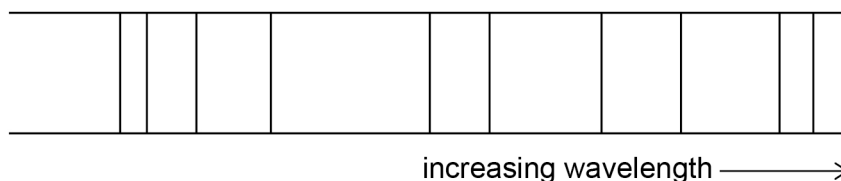
C $1.8 \times 10^8 \text{ m s}^{-1}$

☐

D $3.0 \times 10^8 \text{ m s}^{-1}$

☐

2 4 The diagram shows an atomic line spectrum.



What is the minimum number of energy levels in an atom that could produce this spectrum?

[1 mark]

A 4

☐

B 5

☐

C 9

☐

D 10

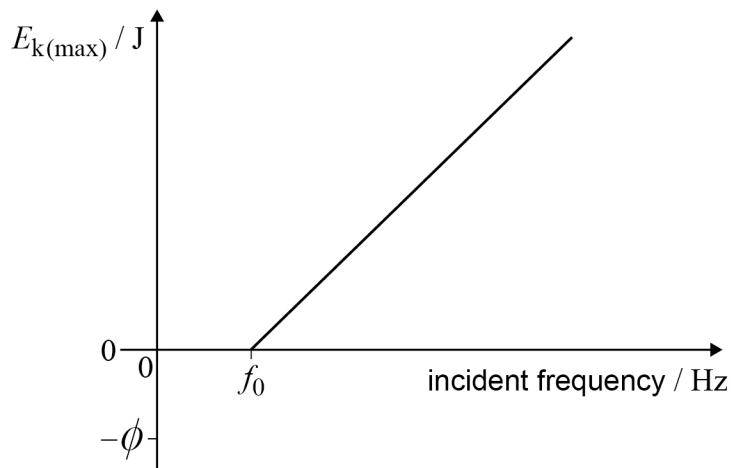
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2 5

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Photoelectrons may be emitted from a metal surface when electromagnetic radiation is incident on the metal surface.

The graph shows the variation of the maximum kinetic energy $E_{k(\max)}$ of the emitted photoelectrons with the frequency of the incident radiation.



Which expression is equivalent to the Planck constant?

[1 mark]

A $\frac{\text{gradient}}{f_0}$

☐

B $\frac{1}{\text{gradient}}$

☐

C $\frac{f_0}{\phi}$

☐

D $\frac{\phi}{f_0}$

☐

[illegible]

[illegible]

[illegible]

[illegible]

Laser light is incident normally on a diffraction grating.

The diffraction grating has 250 slits per millimetre.

A third-order maximum is observed at an angle of 29° to the central maximum.

Calculate the frequency of the laser light.

[4 marks]

frequency = _____ Hz



Don't write

0	2
---	---

Light passing from glass to water has a critical angle of 61° .

0	2
---	---

1

State what is meant by critical angle.

[1 mark]

0	2
---	---

2

Glass has a refractive index of 1.52

Calculate the speed of light in water.

[3 marks]

speed = _____ m s^{-1}

4

Turn over for the next question

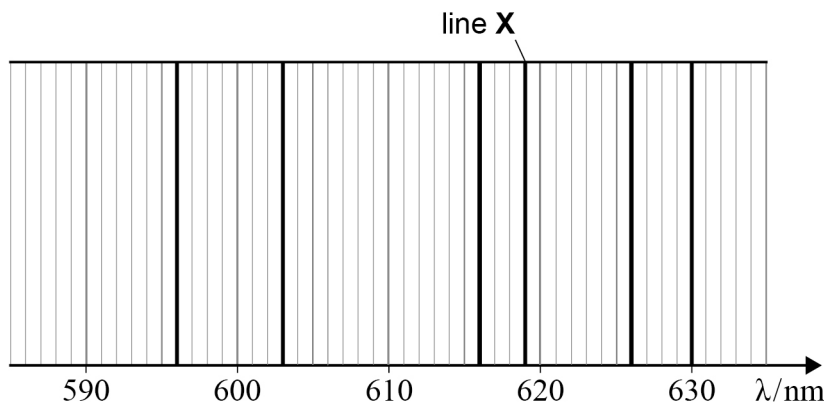
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Figure 3 shows part of the line spectrum for light emitted from a neon lamp. A wavelength scale is shown.

Figure 3



0	4	.	1
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Outline how a diffraction grating can produce a line spectrum.

[2 marks]

0	4	.	2
---	---	---	---

Calculate the energy of the photon responsible for line X in the spectrum in **Figure 3**.

[3 marks]

photon energy = _____ J



Do not write

0 6 Electrons are accelerated from rest through a potential difference V as part of an electron-diffraction experiment.

0 6 . 1 Each electron gains a kinetic energy of 1.5×10^{-15} J.

Calculate V .

[2 marks]

$V =$ _____ V

0 6 . 2 Show that the momentum of an electron after acceleration is approximately 5×10^{-23} kg m s⁻¹.

[3 marks]



0 6 . 3

The electrons are then incident on a graphite crystal. The crystal diffracts the electrons, creating a second-order maximum at an angle of 10° to the zero-order maximum.

The crystal can be modelled as a diffraction grating where the gap between each layer in the crystal behaves like the slit spacing of the grating.

Calculate the effective slit spacing of this diffraction grating.

[4 marks]

slit spacing = _____ m

9

Turn over for the next question

Turn over ►



0 7 . 1 State what is meant by the threshold frequency of radiation in the photoelectric effect. **[2 marks]**

0 7 . 2 Monochromatic light with a photon energy of 3.7×10^{-19} J is incident on a metal surface.

Photoelectrons with a maximum kinetic energy of 5.7×10^{-20} J are emitted from the surface.

Calculate, in eV, the work function of the metal.

[2 marks]

work function = _____ eV



0 7 . 3 The total power of the monochromatic light incident on the surface is 1.3×10^{-2} W.

Calculate the maximum number of photoelectrons that can be emitted from the surface per second.

[2 marks]

maximum number per second = _____

0 7 . 4 The light is replaced with a different monochromatic light with half the wavelength but the same incident power.

Describe and explain any effect on the photoelectrons.

[3 marks]

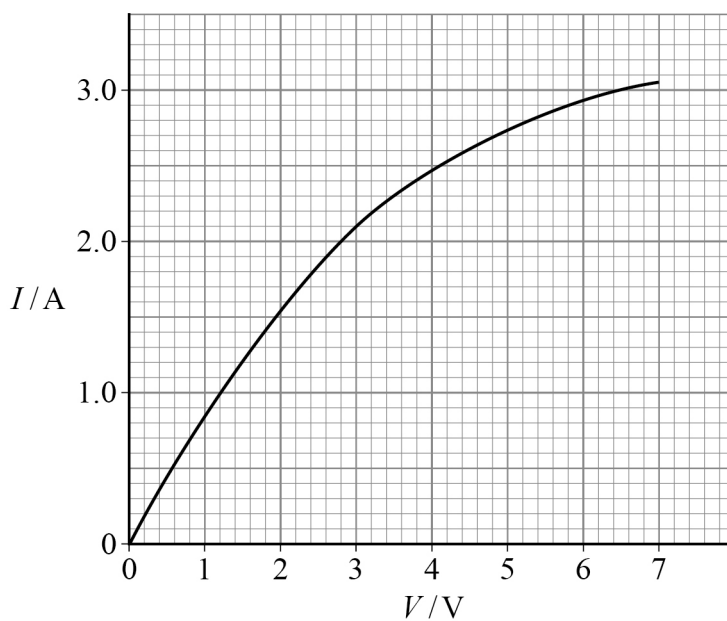


0 8 . 1 Define electrical resistance.

[1 mark]

Figure 6 shows the I - V characteristic of a filament lamp X.

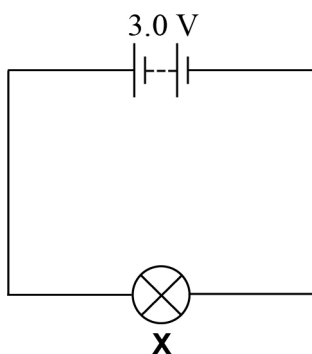
Figure 6



0 8 . 2

Figure 7 shows a battery of emf 3.0 V and negligible internal resistance connected to **X**.

Figure 7



Calculate the resistance of **X** in the circuit shown in **Figure 7**.

[2 marks]

resistance = _____ Ω

0 8 . 3

Calculate the power dissipated by **X** in this circuit.
State the unit for your answer in fundamental (base) units.

[3 marks]

power = _____ unit = _____

Question 8 continues on the next page

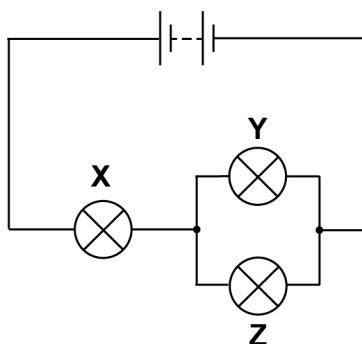
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0 8 . 4

Figure 8 shows **X** connected to two other lamps, **Y** and **Z**.
All three lamps have identical I - V characteristics.

Figure 8



Explain for this circuit how the resistance of **Y** compares with the resistance of **X**.
Calculations are **not** required.

[3 marks]

END OF SECTION A



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ANSWER IN THE SPACES PROVIDED**

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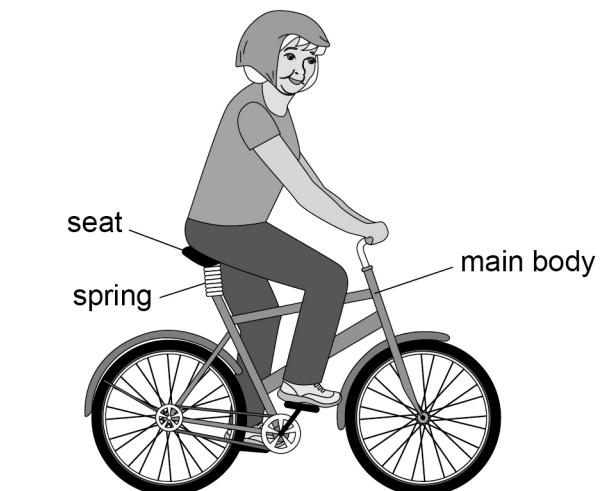
Section B

Answer **all** questions in this section.

0 9

The seat of a bicycle is attached to the main body of the bicycle by a single vertical spring, as shown in **Figure 9**.

Figure 9



The spring has a spring constant of $7.2 \times 10^4 \text{ N m}^{-1}$. The mass of the rider is 68 kg. Assume that the weight of the rider is fully supported by the spring and that the mass of the seat is negligible.

0 9 . 1

Show that the time period for free oscillations of the seat is approximately 0.2 s when the rider is sitting on it.

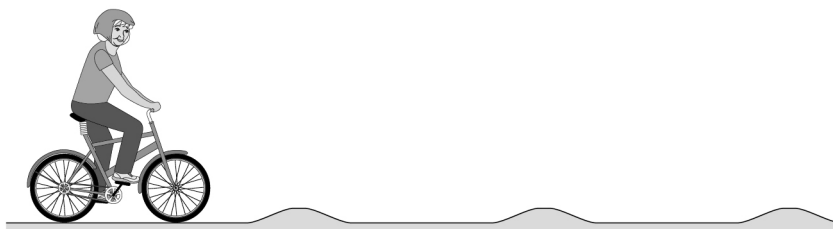
[2 marks]



0 9 . 2

Figure 10 shows the bicycle and rider approaching a series of speed bumps that are equally spaced.

Figure 10



When the bicycle travels over the speed bumps at a certain speed, the rider experiences large-amplitude vertical oscillations.

Identify and explain the effect that causes the large-amplitude oscillations.

[3 marks]

0 9 . 3

The rider experiences large-amplitude oscillations when the bicycle travels at 5.8 m s^{-1} .

Calculate the distance between adjacent speed bumps.

[2 marks]

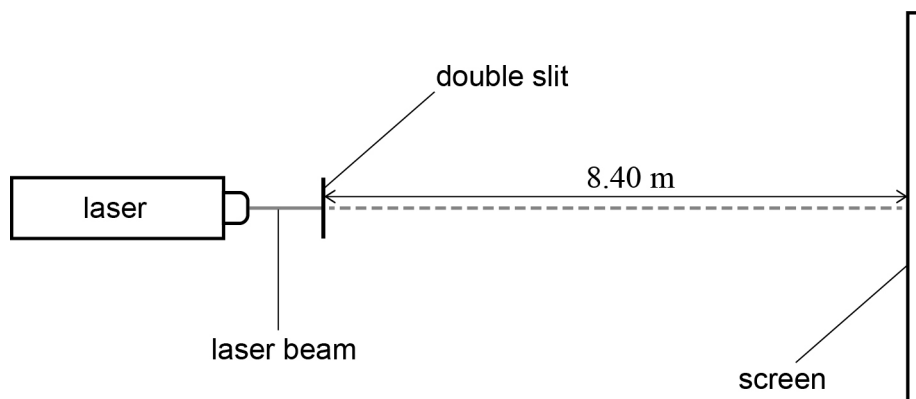
distance = _____ m



1 0

Figure 11 shows apparatus used to determine the wavelength of laser light using Young's double-slit method.

Figure 11



1 0

1

Figure 12 shows the actual size of part of the interference fringe pattern seen on the screen.

Figure 12



Determine, in mm, an accurate measurement for the fringe spacing w .

[2 marks]

$w =$ _____ mm



1 0 . 2 Calculate the percentage uncertainty in your value for w .

[2 marks]

percentage uncertainty = _____

1 0 . 3 The distance between the slits is 0.420 mm with an uncertainty of $\pm 1.2\%$.
The distance from the slits to the screen is 8.40 m with an uncertainty of $\pm 0.6\%$.

Calculate, in nm, the wavelength of the laser light.

[2 marks]

wavelength = _____ nm

1 0 . 4 Calculate the absolute uncertainty in your value for the wavelength.

[3 marks]

absolute uncertainty = _____ nm

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown. 

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown. 

You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional sheets for this working.

1 1

A cell has an emf of 1.5 V and an internal resistance of 2.0 Ω .

A wire of negligible resistance is connected directly from one terminal of the cell to the other.

What is the energy dissipated in the cell in 5 minutes?

[1 mark]

A 5.6 J

☐

B 340 J

☐

C 900 J

☐

D 1400 J

☐


1 2

A metal wire has resistance R .

A second wire made from the same metal is 50% longer but has the same volume as the first wire.

What is the resistance of the second metal wire?

[1 mark]

A $0.67R$

☐

B $1.5R$

☐

C $2.0R$

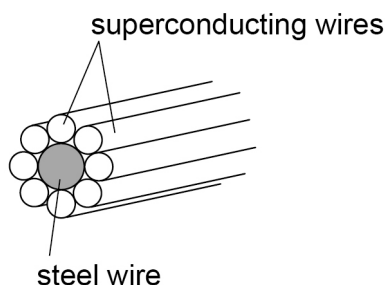
☐

D $2.3R$

☐

1 3

A cable consists of superconducting wires attached in parallel to a steel wire.



What is the purpose of the steel wire in the cable?

[1 mark]

A to increase the critical temperature of the superconductor

☐

B to increase the strength of the cable

☐

C to reduce the resistance of the cable when it is superconducting

☐

D to reduce the current in the cable

☐

1 4

Two resistors **X** and **Y** have resistances R_X and R_Y .

R_Y is greater than R_X .

When **X** and **Y** are combined in parallel, the equivalent resistance is R_T .

Which is correct?

[1 mark]

A $R_T < R_X < R_Y$

☐

B $R_X < R_T < R_Y$

☐

C $R_X < R_Y < R_T$

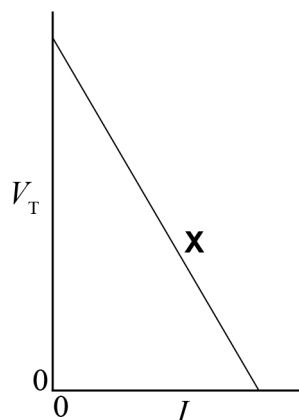
☐

D $R_T < R_Y < R_X$

☐

1 5

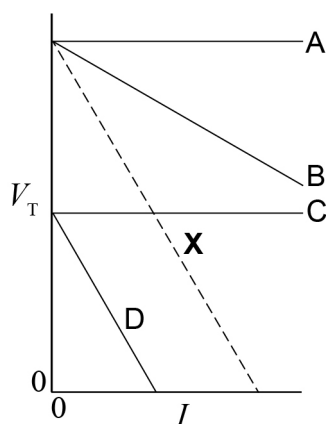
A battery **X** has emf ε and internal resistance r . The graph shows the variation of terminal pd V_T with current I for the battery.



The line for battery **X** is shown again below as a dashed line.

Which line shows the variation of V_T with I for a battery of emf $\frac{\varepsilon}{2}$ and negligible internal resistance?

[1 mark]



A ☐

B ☐

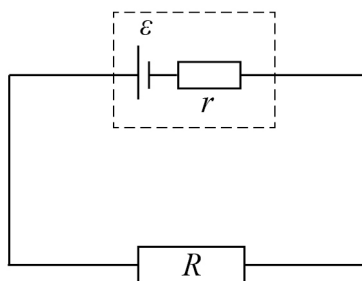
C ☐

D ☐



1 6

A resistor of resistance R is connected to a cell of emf ε and internal resistance r .
The current in the circuit is I .



What is the power dissipated in the resistor of resistance R ?

[1 mark]**A** $I\varepsilon$
☐
B $I^2(R+r)$
☐
C $\frac{\varepsilon^2}{R+r}$
☐
D $\frac{\varepsilon^2 R}{(R+r)^2}$
☐
1 7

Two pendulums have the same length and perform undamped small-angle oscillations with the same amplitude.

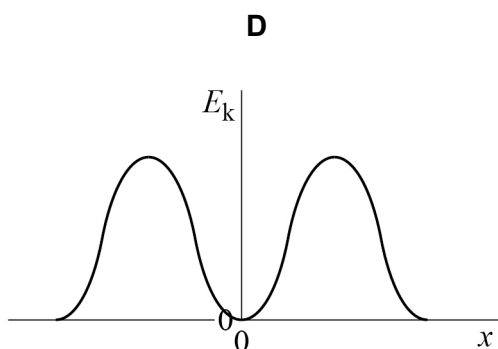
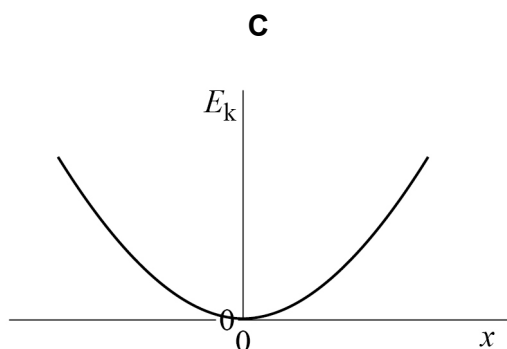
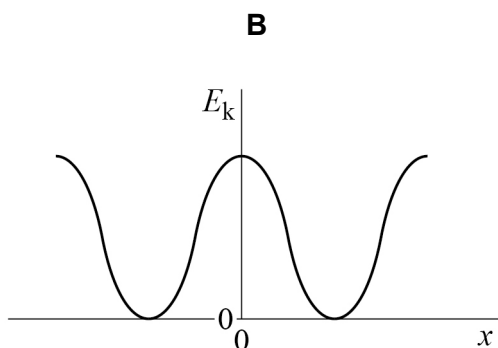
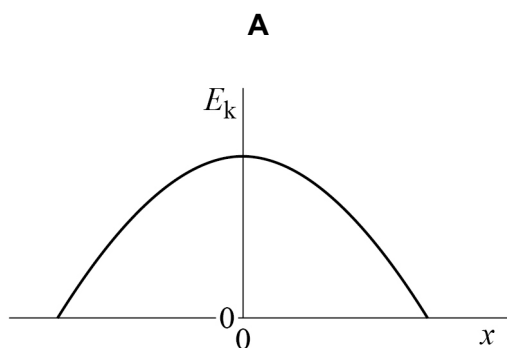
One pendulum bob has a greater mass than the other.

Which quantity is **not** the same for both pendulums?

[1 mark]**A** average speed
☐
B time period
☐
C total energy
☐
D maximum speed
☐


1 8 Which graph shows the variation of kinetic energy E_k with displacement x for a mass–spring system performing simple harmonic motion?

[1 mark]



A ☐

B ☐

C ☐

D ☐

1 9 What **cannot** be observed with sound waves?

[1 mark]

A diffraction ☐

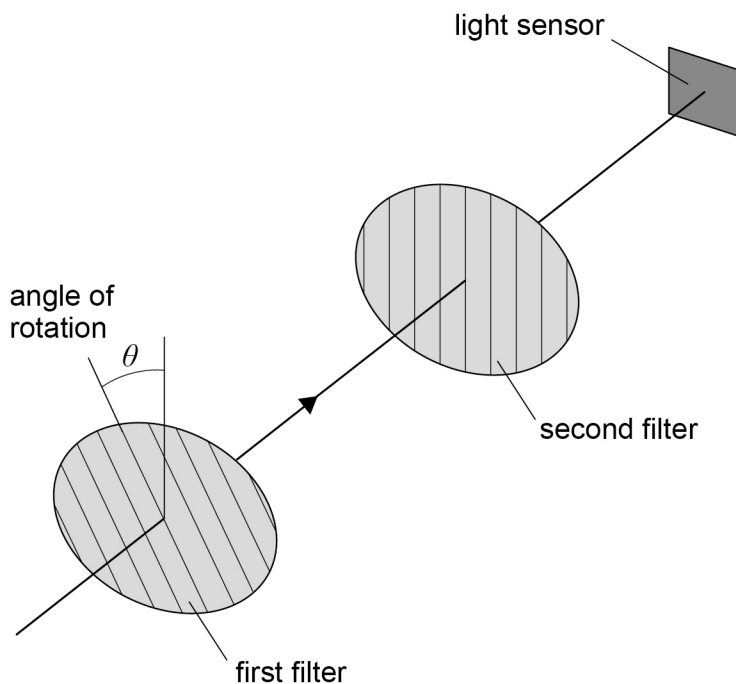
B dispersion ☐

C polarisation ☐

D refraction ☐



Unpolarised light passes perpendicularly through two polarising filters before reaching a sensor that detects its intensity I .



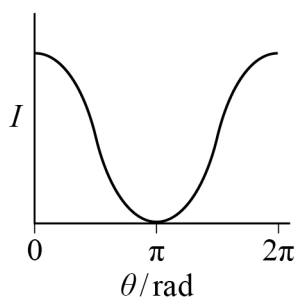
The first filter is rotated through an angle θ about an axis parallel to the light beam.

Which graph on page 29 shows the variation of intensity I with angle θ ?

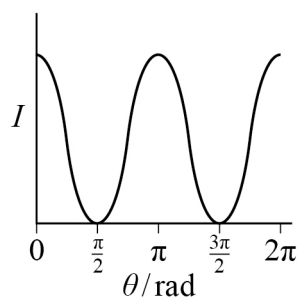
[1 mark]



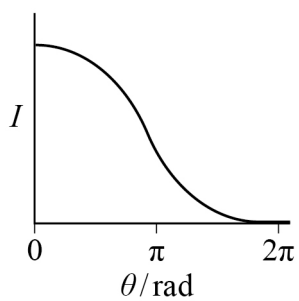
A



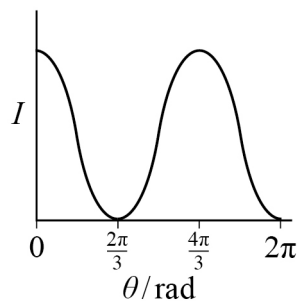
B



C



D



A



B



C



D



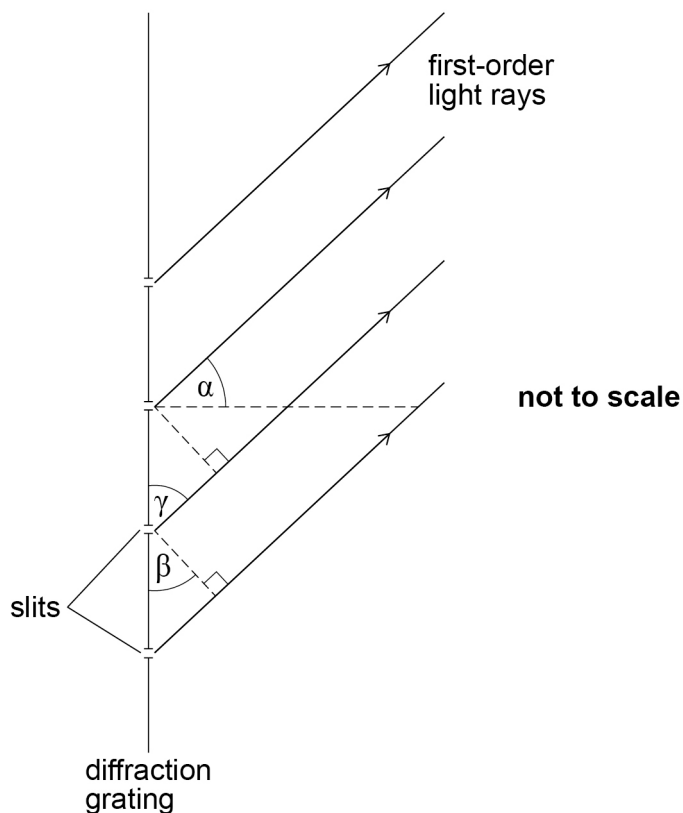
Turn over for the next question

Turn over ►



2 1

In the derivation of the diffraction grating equation $n\lambda = d \sin \theta$, a diagram similar to the one below can be used.



Which of the labelled angles α , β or γ is equivalent to the angle θ in the equation?

[1 mark]

A α only

☐

B α and β only

☐

C γ only

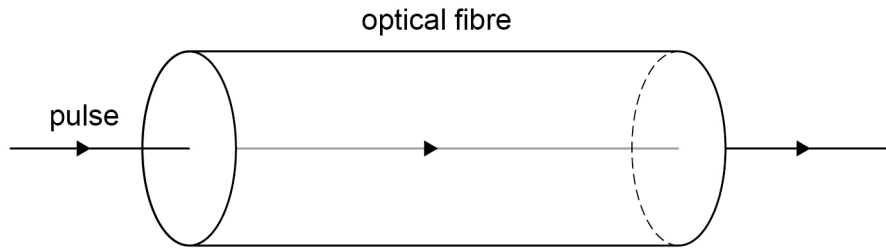
☐

D β and γ only

☐


2 2

A pulse of white light passes along the axis of a straight optical fibre. The pulse is broader when it leaves the optical fibre.



What causes this effect?

[1 mark]

A absorption

☐

B attenuation

☐

C material dispersion

☐

D modal dispersion

☐

2 3

Light crosses the boundary from air into a substance that has a refractive index of 2

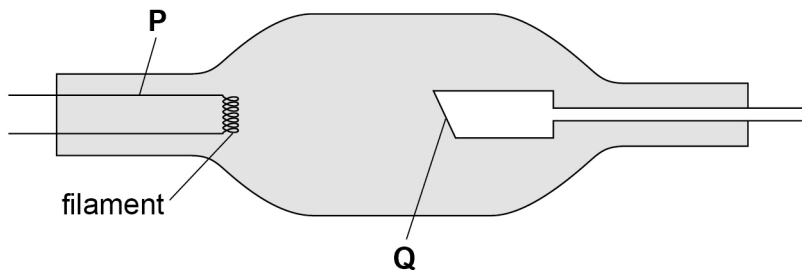
What happens to the speed, frequency and wavelength of the light as it crosses the boundary?

[1 mark]

	Speed	Frequency	Wavelength	
A	halves	stays the same	halves	☐
B	doubles	stays the same	doubles	☐
C	doubles	doubles	stays the same	☐
D	halves	halves	doubles	☐

2 **4** The diagram shows an X-ray tube.

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box



Which row identifies parts **P** and **Q**, and the direction of travel of the electrons?

[1 mark]

	P	Q	Direction of electron travel	
A	anode	cathode	from P to Q	☐
B	anode	cathode	from Q to P	☐
C	cathode	anode	from P to Q	☐
D	cathode	anode	from Q to P	☐

[illegible]





0 3

A narrow single slit of width w is illuminated with light from a laser.
A diffraction pattern is observed on a screen several metres away from the slit.

A total of five maxima is observed on the screen.

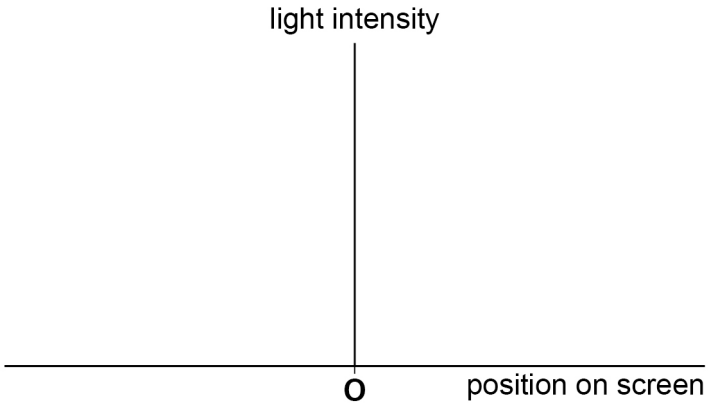
0 3 . 1

Sketch, on **Figure 1**, a graph showing how the light intensity varies with position on the screen.

O marks the central point of the pattern.

[2 marks]

Figure 1



The narrow single slit is replaced with a slit of width greater than w . Nothing else is changed.

A new diffraction pattern is formed.

0 3 . 2

Label, on the position axis of **Figure 1**, the position **M** of the first maximum to the left of **O** for the **new** diffraction pattern.

[1 mark]

0 3 . 3

Describe any change to the intensity of the maxima in the new diffraction pattern.

[1 mark]



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0 5

Photoelectrons are emitted from a sodium surface when it is illuminated with monochromatic electromagnetic radiation.
The stopping potential of these photoelectrons is 3.42 V.

0 5 . 1

Show that the maximum kinetic energy of the photoelectrons is approximately 5.5×10^{-19} J.

[1 mark]

0 5 . 2

Calculate the wavelength of the radiation that is incident on the sodium surface.

work function of sodium = 2.36 eV

[3 marks]

wavelength = _____ m

Question 5 continues on the next page

Turn over ►



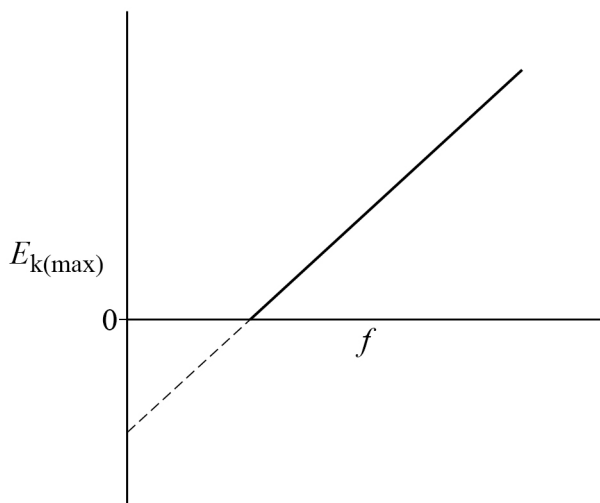
0 5 . 3

The frequency f of the electromagnetic radiation is now varied.

The maximum kinetic energy $E_{k(\max)}$ of the photoelectrons is measured for each value of f .

Figure 2 shows the variation of $E_{k(\max)}$ with f .

Figure 2



Magnesium and caesium surfaces are also illuminated with electromagnetic radiation and photoelectrons are emitted from the surfaces.

Draw, on **Figure 2**, lines to show the variation of $E_{k(\max)}$ with f for magnesium **and** for caesium.

Label your lines.

work function of magnesium = 3.68 eV

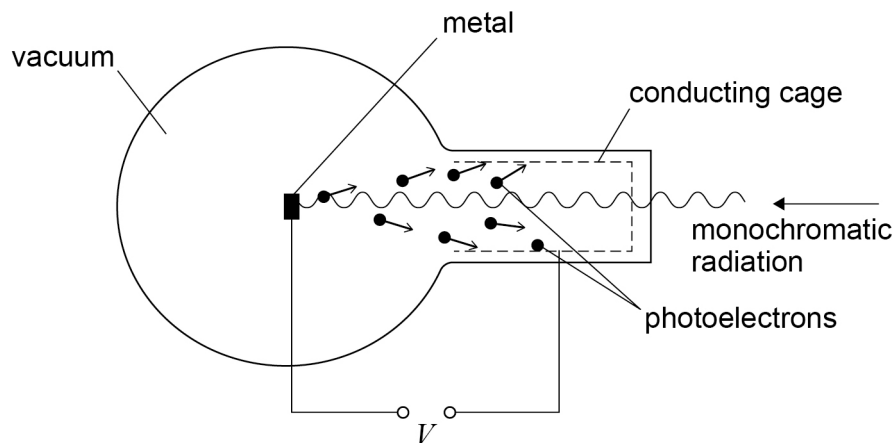
work function of caesium = 2.14 eV

[2 marks]



0 5 . 4 Figure 3 shows apparatus that can be used to investigate the photoelectric effect.

Figure 3



The experiment is conducted in a vacuum.

The photoelectrons that are produced move towards the conducting cage. The stopping potential V of the photoelectrons is measured using a circuit connected to the metal and the conducting cage.

A small amount of air now leaks into the apparatus.

Explain how the measured value of V is affected by the presence of the air.

[2 marks]

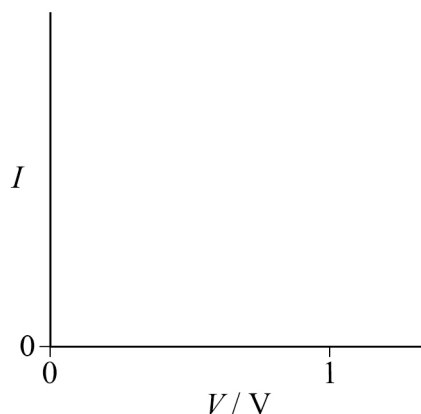
06.1

A diode conducts only when connected the correct way round in a circuit.

Sketch, on **Figure 4**, a graph of the variation of current I with potential difference V over a range of 0 to +1 V for a diode that becomes conducting.

[1 mark]

Figure 4

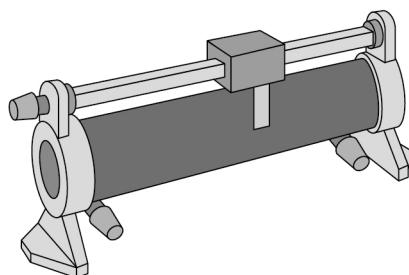


A student wants to obtain data to be able to draw the graph in **Figure 4**.

The student has a 1.5 V cell, the diode, an ammeter, a voltmeter, connecting wires and a rheostat like that shown in **Figure 5**.

The rheostat can be used as a variable resistor or as a potential divider.

Figure 5



0 6 . 2

Draw a suitable circuit in which the rheostat is used as a variable resistor.

[2 marks]

0 6 . 3

Draw a suitable circuit in which the rheostat is used as a potential divider.

[1 mark]

0 6 . 4

Explain the advantage of using the rheostat as a potential divider rather than as a variable resistor.

[1 mark]



A student builds a circuit that uses a filament lamp to heat a tank for reptiles. The circuit includes components that are designed to control the temperature of the tank. A thermistor and the lamp are positioned inside the tank.

Figure 6 shows the circuit. The battery has negligible internal resistance.

Figure 7A shows the variation of resistance R_{th} with temperature T for the thermistor.

Figure 7B shows the variation of potential difference V_L with current I for the lamp.

Figure 6

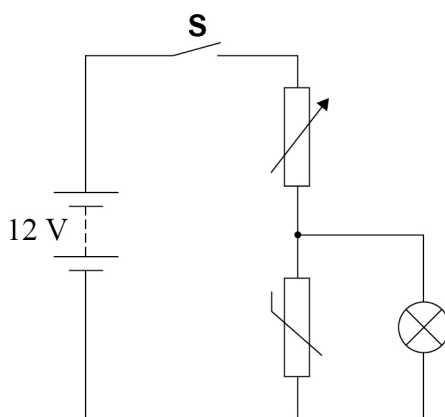


Figure 7A

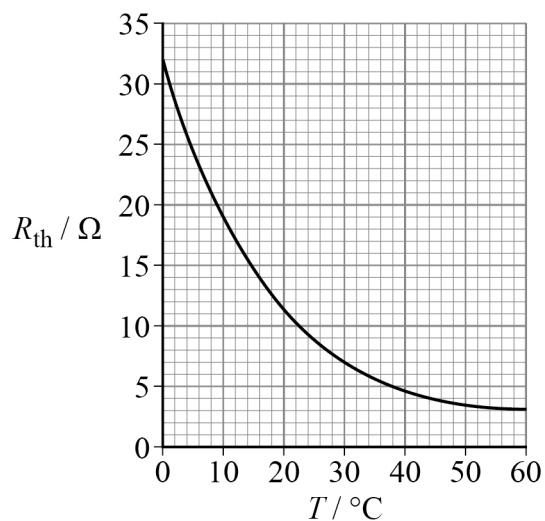
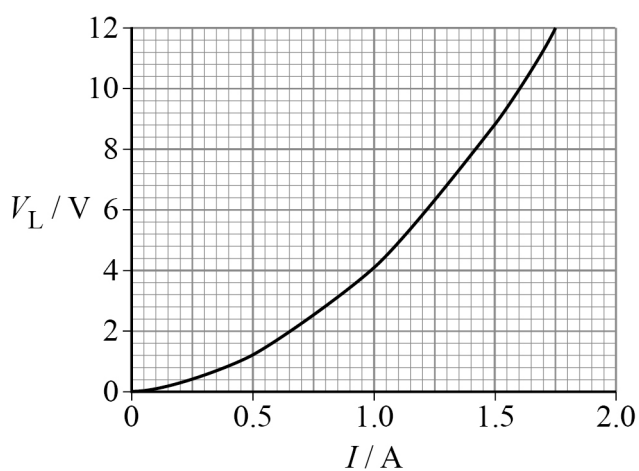


Figure 7B



The initial temperature in the tank is $20\text{ }^{\circ}\text{C}$.

07.1

Switch **S** is closed and the potential difference across the lamp is 3.0 V .

Show that the resistance of the parallel combination of the thermistor and lamp immediately after **S** is closed is approximately $2.7\text{ }\Omega$.

[4 marks]

07.2

Determine the resistance of the variable resistor.

[2 marks]

resistance = _____ Ω

07.3

Determine the power of the lamp immediately after **S** is closed.

[1 mark]

power = _____ W

Question 7 continues on the next page

Turn over ►



0 7 . 4

Suggest how the resistance of the parallel combination of the lamp and thermistor is likely to change in the period immediately after **S** is closed.

[2 marks]

0 7 . 5

On another occasion the initial temperature in the tank is again 20 °C.
This time, the resistance of the variable resistor in **Figure 6** is increased before **S** is closed.

State and explain the effect of this change on the initial rate of temperature increase in the tank.

[2 marks]



Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

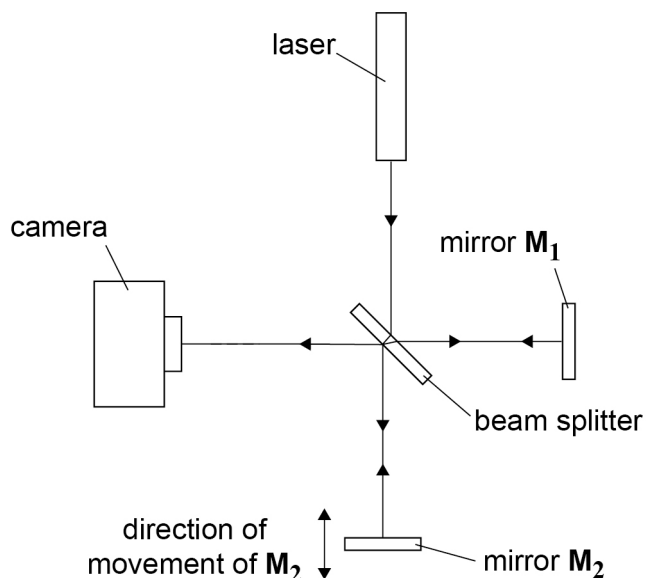
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08

The interferometer shown in **Figure 8** uses interference to measure very small movements of the mirror M_2 .

Figure 8



Monochromatic light from a laser arrives at a beam splitter.

Half of the light is reflected to mirror M_1 . The other half of the light goes through the beam splitter and on to mirror M_2 .

Light is reflected from M_1 and M_2 and travels via the beam splitter to a camera.

Light from M_1 interferes with light from M_2 at the camera. The effects of the interference are observed.

When M_2 moves, the interference observed at the camera changes. This enables a determination of the distance that M_2 moves.

08

. 1

Explain why M_1 and M_2 act as coherent sources.

[2 marks]



Mirror M_2 is moved a distance d to a new position closer to the beam splitter.
As M_2 moves, light entering the camera changes from maximum to minimum intensity.

0 8 . 2

Explain the change in light intensity.

[2 marks]

0 8 . 3

Explain, using an appropriate calculation, why the minimum value of d is approximately 1.6×10^{-7} m.

The wavelength of light from the laser is 650 nm.

[2 marks]

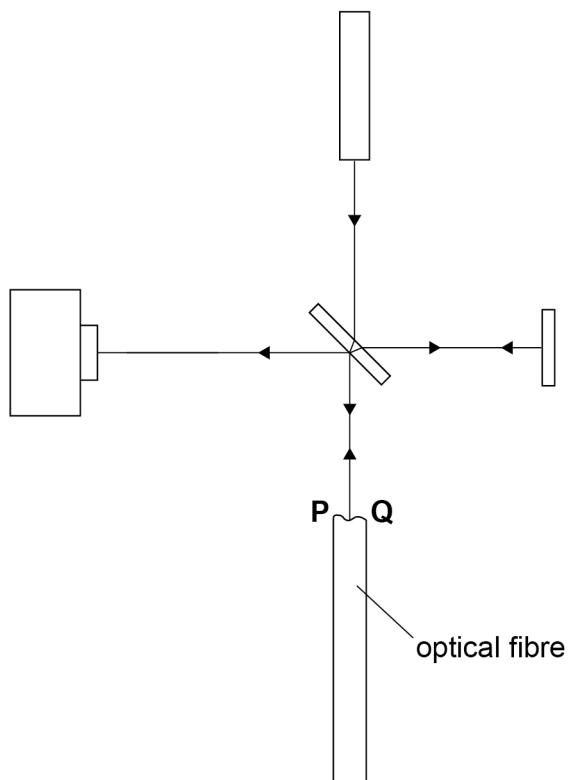
Question 8 continues on the next page



0 8 . 4 The interferometer is used to examine the end of an optical fibre.

Mirror M_2 is replaced with the end of an optical fibre. The end of the optical fibre reflects light back to the beam splitter as shown in **Figure 9**.

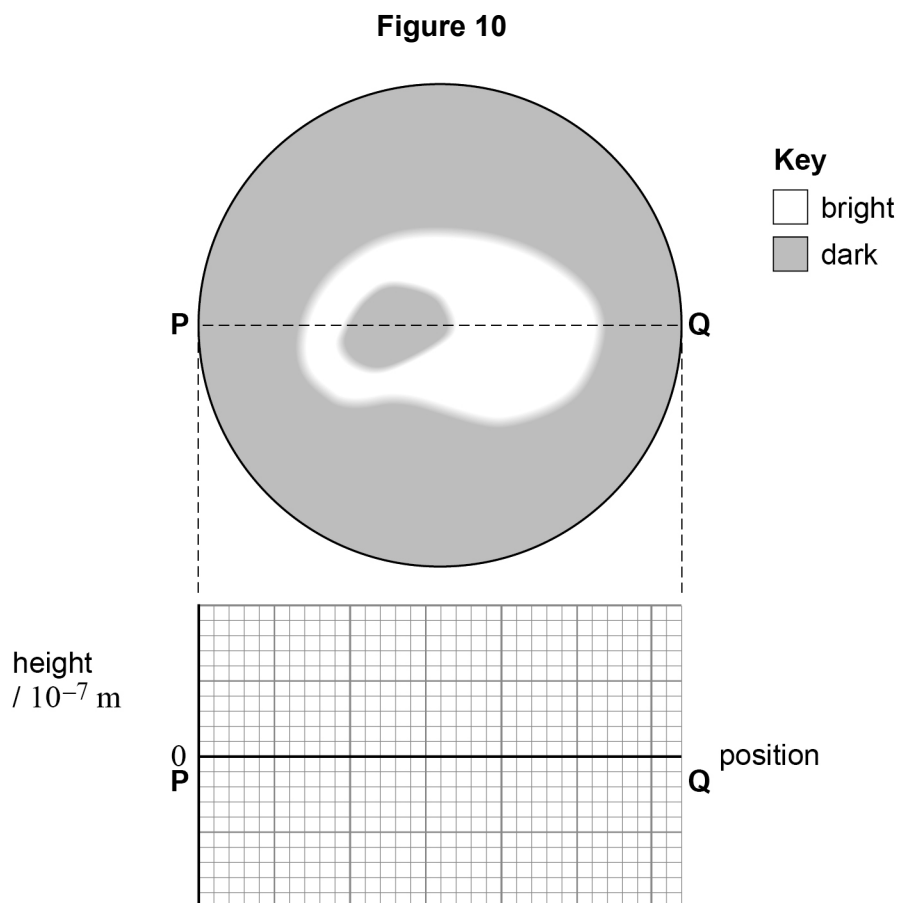
Figure 9



The end of the optical fibre between **P** and **Q** is uneven because it varies in height.



Figure 10 shows the interference pattern of light observed at the camera.
PQ is a diameter of the circular end of the optical fibre.



Draw, on **Figure 10**, a graph showing the variation of height with position along **PQ**.

Add a suitable scale to your height axis.

[3 marks]

Question 8 continues on the next page



0 8 . 5 A different laser is now used. This laser emits light with a smaller wavelength.

Suggest how this change affects the resolution of the interferometer.

[1 mark]

0 8 . 6 The optical fibre with an uneven end is used to transmit information.

Explain why the uneven end of the optical fibre increases modal dispersion.

[2 marks]

12

END OF SECTION A



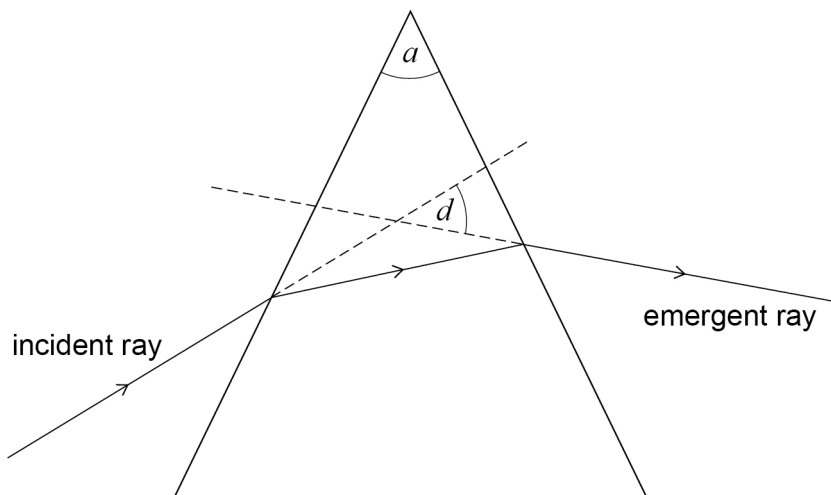
Section B

Answer **all** questions in this section.

0	9
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Figure 11 shows a ray of monochromatic light passing through a glass prism.

Figure 11



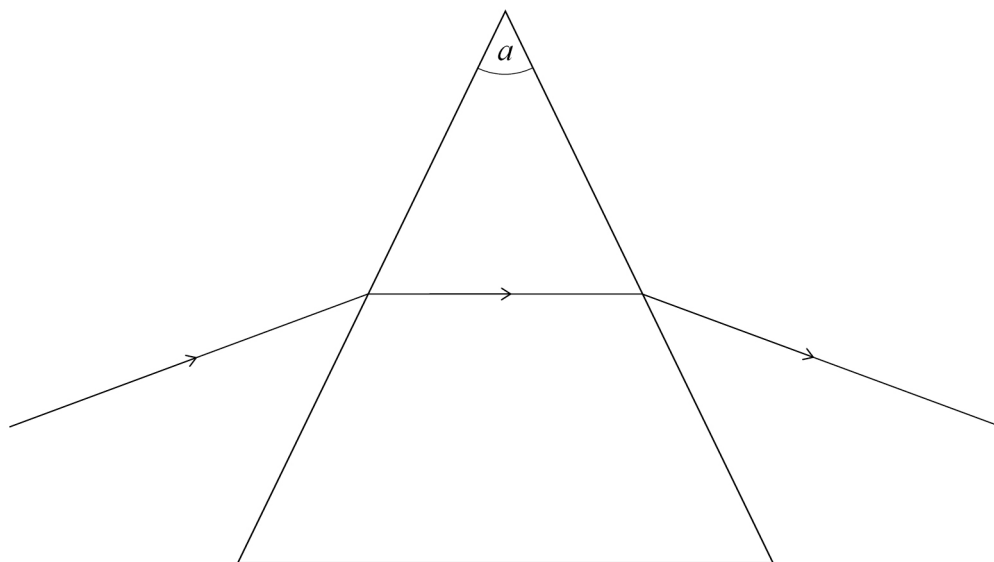
The angle a is the refracting angle of the prism.

The angle of deviation d of the ray is the angle between the direction of the incident ray and the direction of the emergent ray as shown on **Figure 11**.

Question 9 continues on the next page

09.1 Figure 12 shows a ray passing through a prism on a symmetrical path.

Figure 12



this diagram is drawn to scale

Measure d for this light ray.

[2 marks]

$d =$ _____ °

09.2 Estimate the absolute uncertainty in your value for d .

[1 mark]

absolute uncertainty in $d =$ _____ °

09.3 Explain the reasoning for your answer to Question **09.2**.

[1 mark]



For the ray diagram in **Figure 12**, the refractive index n of the glass in the prism is given by

$$n = \frac{\sin \frac{(a+d)}{2}}{\sin \left(\frac{a}{2} \right)}$$

0 9 . 4 In **Figure 12**, $a = (50.0 \pm 0.5)^\circ$.

Calculate $\frac{(a+d)}{2}$ and estimate its percentage uncertainty.

[2 marks]

$$\frac{(a+d)}{2} = \underline{\hspace{2cm}}^\circ$$

$$\text{percentage uncertainty} = \underline{\hspace{2cm}}$$

0 9 . 5 Calculate n and estimate its percentage uncertainty.

Assume that the percentage uncertainty in the sine of the angles is the same as the percentage uncertainty in the angles themselves.

[2 marks]

$$n = \underline{\hspace{2cm}}$$

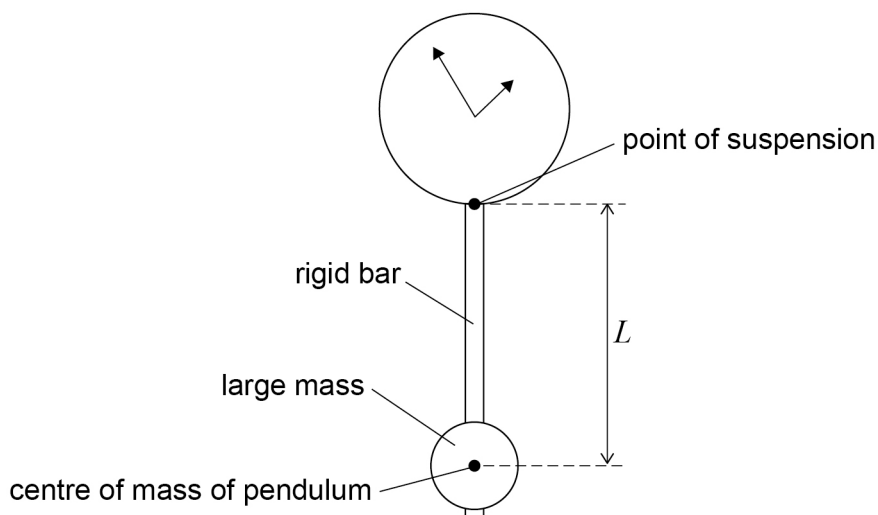
$$\text{percentage uncertainty} = \underline{\hspace{2cm}}$$

1 0

Pendulums are used to make some clocks keep the correct time.

Figure 13 shows a clock with a pendulum. The pendulum is made from a rigid bar of negligible mass with a large mass near the lower end.

Figure 13



Assume that the pendulum behaves as a simple pendulum with a length equal to the distance L between the point of suspension and the centre of mass of the pendulum.

The pendulum must oscillate with a time period of 1.400 s for the clock to keep correct time.

1 0

. 1

The pendulum oscillates with a period of 1.402 s.

Calculate the error in the time measured by the clock in 24 hours.

[3 marks]

error = _____ s



1 0 . 2 Show that L is approximately 0.5 m.

[2 marks]

1 0 . 3 The owner of the clock wants to make the clock more accurate by attaching a small additional mass to the pendulum.

Explain where the owner should attach the small additional mass.

[2 marks]

1 0 . 4 The clock is moved to a warmer room, causing the rigid bar to increase in length.

Explain how the owner must move the small additional mass in order for the clock to keep accurate time.

[1 mark]

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown. 

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown. 

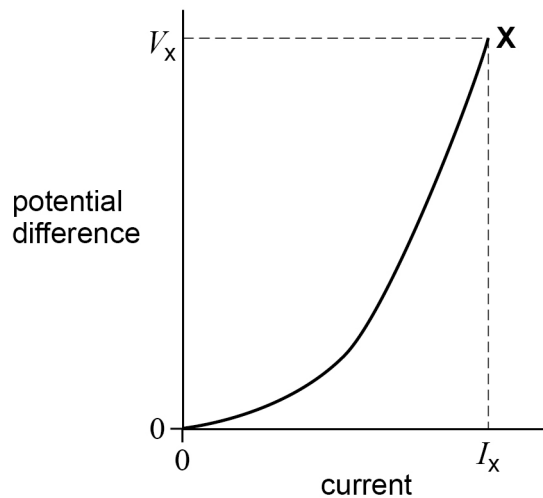
You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

1 1 Which row correctly relates a type of wave to its direction of oscillation and its polarisation?
[1 mark]

	Type of wave	Direction of oscillation	Polarisation	
A	transverse	parallel to the direction of energy transfer	can be polarised	☐
B	transverse	perpendicular to the direction of energy transfer	cannot be polarised	☐
C	longitudinal	parallel to the direction of energy transfer	cannot be polarised	☐
D	longitudinal	perpendicular to the direction of energy transfer	can be polarised	☐



1 2 The graph shows the variation of potential difference V with current I for a device.



Which row shows the resistance of the device at point **X** and the power transferred in the device at point **X**?

[1 mark]

	Resistance	Power transferred	
A	the gradient at X	the area under the curve	☐
B	the gradient at X	$V_x I_x$	☐
C	$\frac{V_x}{I_x}$	the area under the curve	☐
D	$\frac{V_x}{I_x}$	$V_x I_x$	☐

Turn over for the next question

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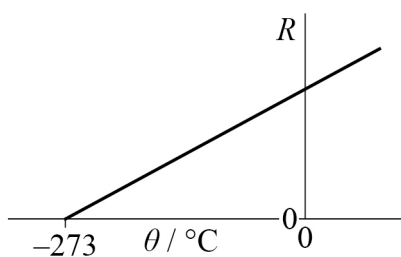


1 3

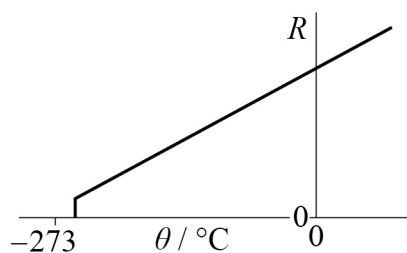
Which graph shows the variation of resistance R with Celsius temperature θ for a superconductor?

[1 mark]

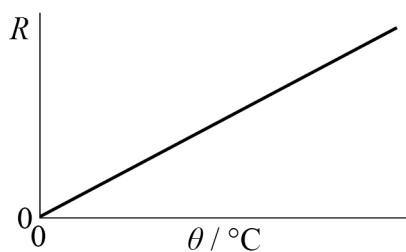
A



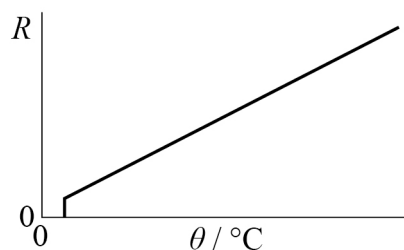
B



C



D



A

☐

B

☐

C

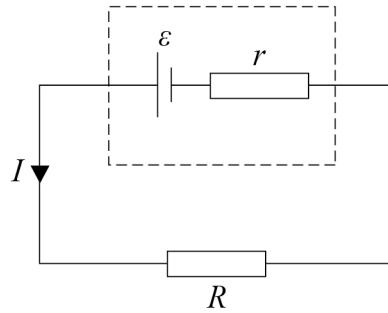
☐

D

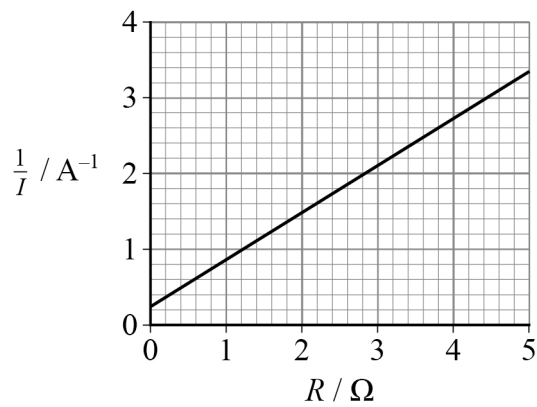
☐


1 4

A cell of emf ε and internal resistance r is connected to an external resistor.
The current I is measured for a range of values of R , the resistance of the external resistor.



The graph shows the variation of $\frac{1}{I}$ with R .



What are the values of ε and r ?

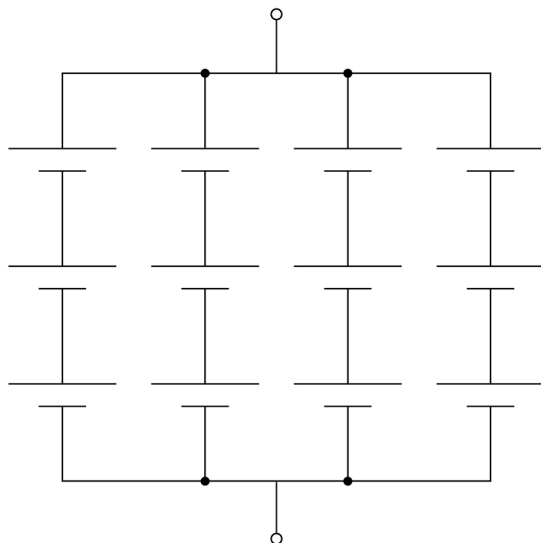
[1 mark]

	ε / V	r / Ω
A	0.63	0.16
B	0.63	0.40
C	1.6	0.16
D	1.6	0.40

☐
☐
☐
☐

1 5

Twelve identical cells are placed in a network.
Each cell has an emf ε and internal resistance r .



What are the emf and the internal resistance of the network?

[1 mark]

	Emf	Internal resistance
A	3ε	$\frac{3}{4}r$
B	3ε	$\frac{4}{3}r$
C	4ε	$\frac{4}{3}r$
D	4ε	$\frac{3}{4}r$

☐
☐
☐
☐


- 1 6** A mass–spring system oscillates with a period of T .
The spring constant is k and the mass is m .
A second mass–spring system has a spring constant of $3k$ and a mass of $5m$.

What is the period of oscillation of the second system?

[1 mark]

A $0.36T$

☐

B $0.78T$

☐

C $1.3T$

☐

D $2.8T$

☐

- 1 7** A mass–spring system will resonate only when the forcing oscillation has:

[1 mark]

A an amplitude that is any multiple of the amplitude of the resonating system.

☐

B an amplitude that is equal to the amplitude of the resonating system.

☐

C a frequency that is any multiple of the natural frequency of the resonating system.

☐

D a frequency that is equal to the natural frequency of the resonating system.

☐

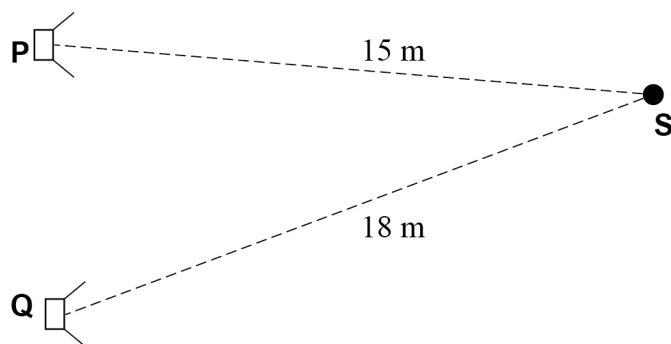
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1 8

Sound from two loudspeakers **P** and **Q** is emitted in antiphase.
P and **Q** produce sound of frequency 510 Hz.



The speed of sound is 340 m s^{-1} .
S is 15 m from **P** and 18 m from **Q**.

When observed at **S**, the sound from **P** is:

[1 mark]

A in phase with the sound from **Q**.

☐

B in antiphase with the sound from **Q**.

☐

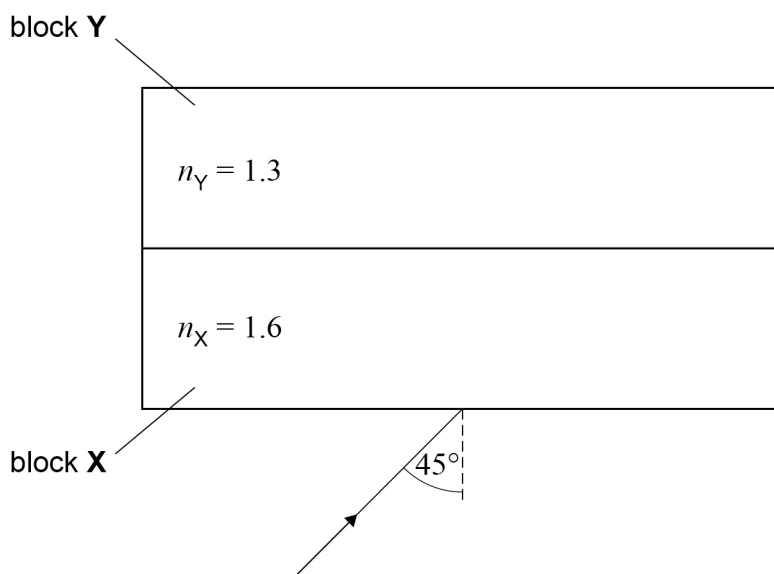
C $\frac{\pi}{2}$ rad ahead of the sound from **Q**.

☐

D $\frac{\pi}{4}$ rad ahead of the sound from **Q**.

☐


Questions 19 and 20 relate to two parallel-sided blocks **X** and **Y** joined together as shown.



A ray of light enters block **X** with an angle of incidence of 45° . The ray passes through the blocks and emerges from the top surface of block **Y**. The refractive indices for block **X** and block **Y** are shown on the diagram.

1 9

The ratio $\frac{\text{speed of light in block X}}{\text{speed of light in block Y}}$ is

[1 mark]

A 0.20

☐

B 0.81

☐

C 1.2

☐

D 2.8

☐

2 0

The ray emerges from the top surface of block **Y** at an angle of θ to the normal.

Which is correct?

[1 mark]

A $\theta = 90^\circ$

☐

B $\theta > 45^\circ$

☐

C $\theta = 45^\circ$

☐

D $\theta < 45^\circ$

☐

2 1 Which is true about fluorescent tubes?

[1 mark]

- A** An electric current in the tube ionises the phosphor.
- B** Phosphor atoms emit only UV photons as they de-excite.
- C** Mercury ions become excited by collisions in the tube.
- D** Mercury ions emit only visible photons as they de-excite.

☐
☐
☐
☐

2 2 The diagram shows three of the energy levels, **O**, **P** and **Q**, in an atom.

_____ **O** = 0 J

_____ **P** = -5.4×10^{-19} J

not to scale

_____ **Q** = -22×10^{-19} J

A photon is emitted during a transition between two energy levels.

Which row correctly identifies the transition and the frequency of the emitted photon?

[1 mark]

	Transition	Frequency of emitted photon / 10^{15} Hz	
A	Q to O	3.3	☐
B	P to O	8.1	☐
C	O to Q	3.3	☐
D	O to P	8.1	☐



2 3 The photoelectric effect shows that when photoelectrons are emitted:

[1 mark]

- A** the atoms have accumulated sufficient energy gradually.
- B** energy is always absorbed in discrete amounts.
- C** the photoelectrons can only have certain permitted kinetic energies.
- D** the photoelectrons show wave-like properties as they are emitted.

☐
☐
☐
☐

2 4 The momentum p of a particle is given by $p = \sqrt{2E_k m}$ where E_k is the kinetic energy and m is the mass of the particle.

An electron and a proton are given the same kinetic energy.

What is the ratio $\frac{\text{de Broglie wavelength of the electron}}{\text{de Broglie wavelength of the proton}}$?

[1 mark]

A 2.3×10^{-2}

☐

B 4.3×10^1

☐

C 1.8×10^3

☐

D 3.4×10^6

☐

[illegible]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

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2 1 1 X P H 0 2

A diffraction grating of width 42 mm has 12 000 lines.
Monochromatic light is incident normally on the grating.
The angle between the two second-order diffraction maxima is 41.4° .

Calculate the wavelength of the incident light.

[4 marks]

wavelength = _____ m

4

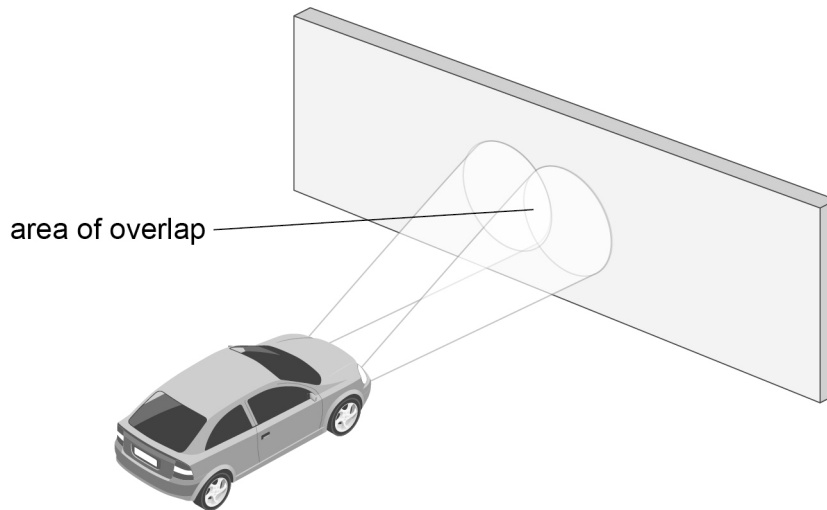
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Do not write Do not write

Light from the filament lamps in a car's headlights is incident on a white wall. The two beams overlap on the white wall as shown in **Figure 1**.

Figure 1



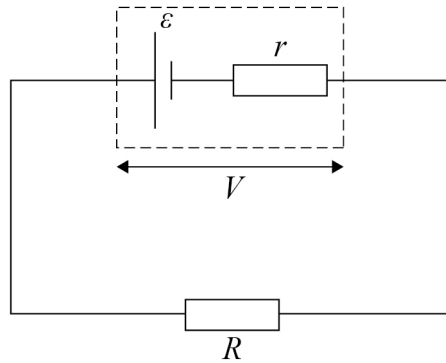
Explain why no interference pattern is observable to the naked eye in the area of overlap.

[3 marks]

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Figure 1 shows a cell connected to an external resistor of resistance R . The internal resistance of the cell is r .

Figure 1



Explain what is meant by the quantities represented by V and ε in **Figure 1**.

[3 marks]

Turn over for the next question



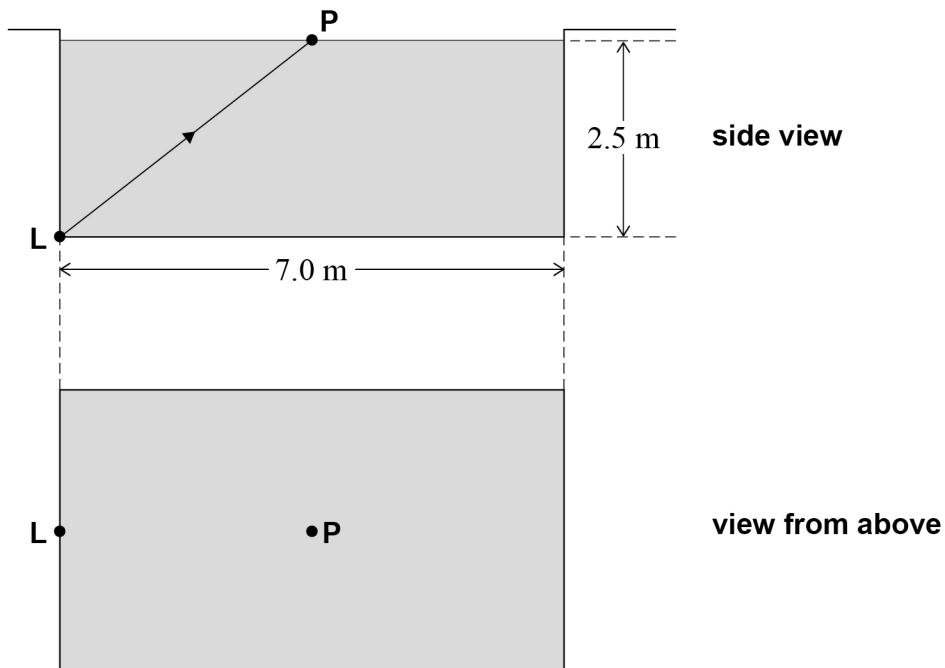
0 3 . 1 Show that the critical angle for light travelling from water to air is approximately 49° .

speed of light in water = $2.25 \times 10^8 \text{ m s}^{-1}$

[2 marks]

Figure 2 shows a pool of water with a lamp at position **L**.
L is at the bottom of the pool, halfway across the pool at one end.
P is a point on the surface of the water at the centre of the pool.
 The pool is 7.0 m long and the water is 2.5 m deep.
 The surface of the water is smooth.

Figure 2



0 3 . 2

State and explain whether a ray of light from **L** can emerge from the water at **P**.
Support your answer with a calculation.

[2 marks]

Question 3 continues on the next page

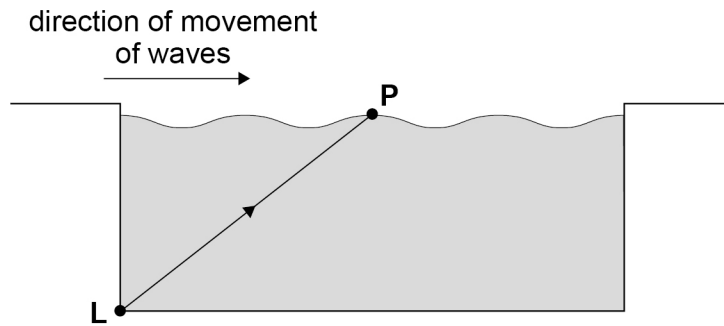
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0 3 . 3

Figure 3 shows the same pool with small waves moving continuously along the surface from left to right.

Figure 3



Suggest, using one or more diagrams, whether a ray of light from **L** can emerge from the water at **P** as the waves move.

[3 marks]



0 4 . 1 State what is meant by monochromatic light.

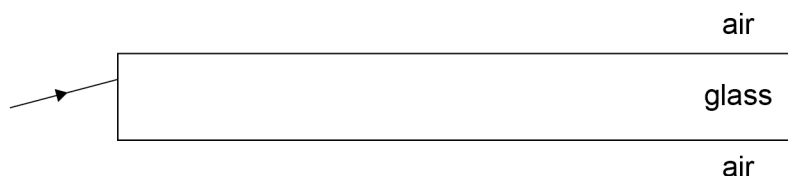
[1 mark]

0 4 . 2 Explain the benefit of using monochromatic light in optical fibres.

[2 marks]

0 4 . 3 **Figure 4** shows a simple optical fibre consisting of a thin cylinder of glass in air.

Figure 4



Explain how changes to this design can reduce modal dispersion.

[3 marks]



Do not write

2

0

What does **not** contribute to the formation of a spectrum when light passes through a diffraction grating?

[1 mark]

A destructive interference

☐

B diffraction at the slits

☐

C constructive interference

☐

D polarisation at the slits

☐

Don't write

2

1

A diffraction grating has a spacing of $100 \text{ lines mm}^{-1}$.

Light of wavelength $6.33 \times 10^{-7} \text{ m}$ is incident normally on the grating.

What is the angle between the two second-order diffraction maxima?

[1 mark]

A 3.6°

☐

B 7.3°

☐

C 14.5°

☐

D 29.1°

☐

The critical angle for light travelling from diamond to air is 24° .
The critical angle for light travelling from glass to air is 42° .

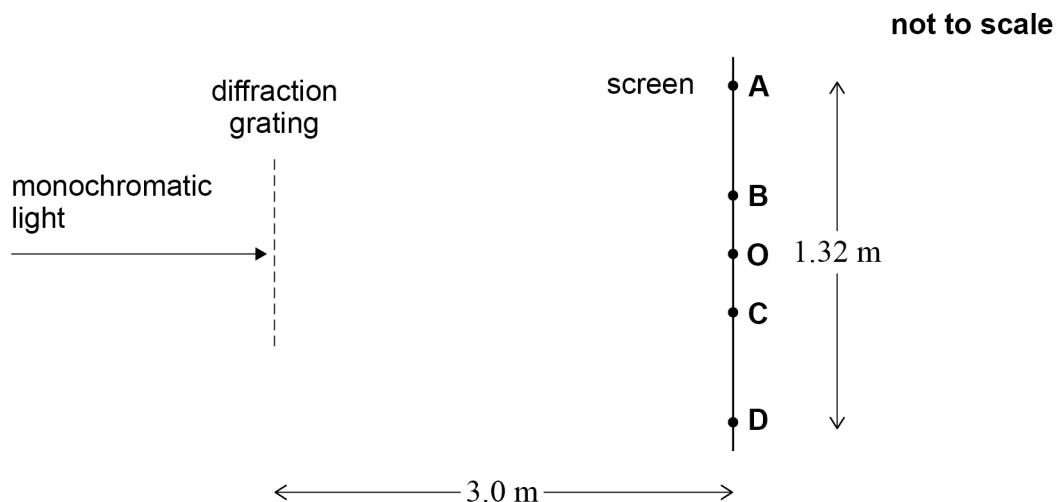
Explain what can be deduced about the relative values of the refractive index of diamond and the refractive index of glass.

[2 marks]

09

A screen is placed 3.0 m from a diffraction grating as shown in **Figure 10**. The screen and grating are parallel.

Figure 10



Monochromatic light of wavelength 540 nm is incident normally on the diffraction grating. The central maximum appears on the screen at **O**. The first-order and second-order maxima on each side of **O** are seen at positions **A**, **B**, **C** and **D**.

The distance between **A** and **D** is 1.32 m.

09

.1

Show that the angle between the two second-order maxima is approximately 25° .

[2 marks]



0 9 . 2

Calculate the number of lines per mm on the grating.

[3 marks]

number of lines per mm = _____

0 9 . 3

The incident light is now replaced with monochromatic light of a different wavelength.

More maxima are observed on the screen.

State and explain any other change in the appearance of the pattern on the screen.

[3 marks]

END OF SECTION A



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Section B

Answer **all** questions in this section.

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1**5**

The cladding in optical fibres

[1 mark]

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A has a higher refractive index than the core.



B reduces material dispersion inside the core.



C increases modal dispersion inside the core.



D increases the critical angle at the surface of the core.



0

7

4

An accurate determination of y can be made by analysing reflections **R** and **S**.

Show that the path difference between **R** and **S** is independent of d .

[2 marks]



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4

Light of wavelength 600 nm and light of an unknown wavelength λ are both incident normally on a diffraction grating.

The angle of the fourth-order maximum for the light of wavelength 600 nm equals the angle of the fifth-order maximum for the light of wavelength λ .

What is λ ?

[1 mark]

A 450 nm

☐

B 480 nm

☐

C 680 nm

☐

D 750 nm

☐

1 7

Which combination of the refractive index of the core and the refractive index of the cladding causes the smallest modal dispersion in an optical fibre?

[1 mark]

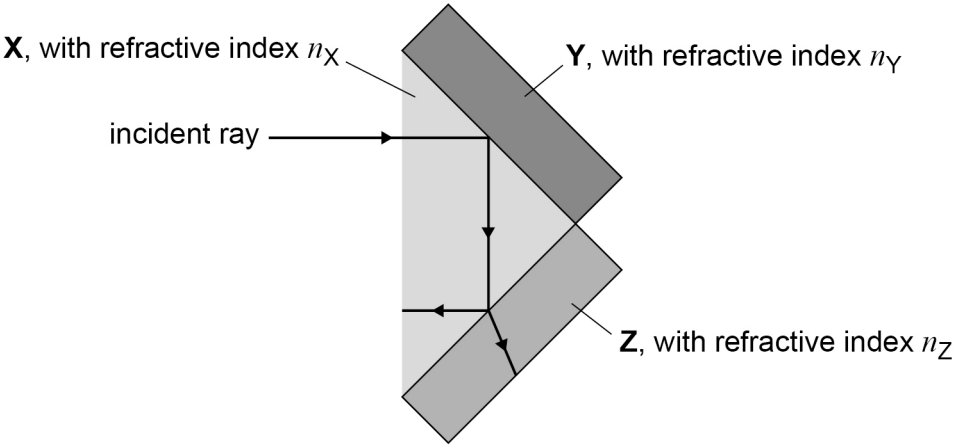
	Refractive index of core	Refractive index of cladding	
A	1.2	1.4	☐
B	1.4	1.2	☐
C	2.2	2.4	☐
D	2.4	2.2	☐



Don't write

A triangular prism made of substance **X** is joined to blocks made of substances **Y** and **Z**.

The diagram shows the path of a ray of light.



Total internal reflection occurs at the **XY** boundary.

Which row is correct?

[1 mark]

A	$n_X > n_Y$	$n_Y > n_Z$	☐
B	$n_X > n_Y$	$n_Y < n_Z$	☐
C	$n_X < n_Y$	$n_Y < n_Z$	☐
D	$n_X < n_Y$	$n_Y > n_Z$	☐

2**3**

Which is **not** a necessary assumption in deriving the diffraction grating equation $n\lambda = d\sin\theta$?

[1 mark]

A Maxima are produced by path differences of integer wavelengths.

☐

B Diffraction occurs at the grating slits.

☐

C The grating slits are separated by a constant distance.

☐

D The light from the grating must be polarised.

☐



0 8 . 1

Explain what is meant by modal dispersion in an optical fibre.

[2 marks]

The speed of light in the glass core of an optical fibre is $1.97 \times 10^8 \text{ m s}^{-1}$.

0 8 . 2

Calculate the refractive index of the glass.

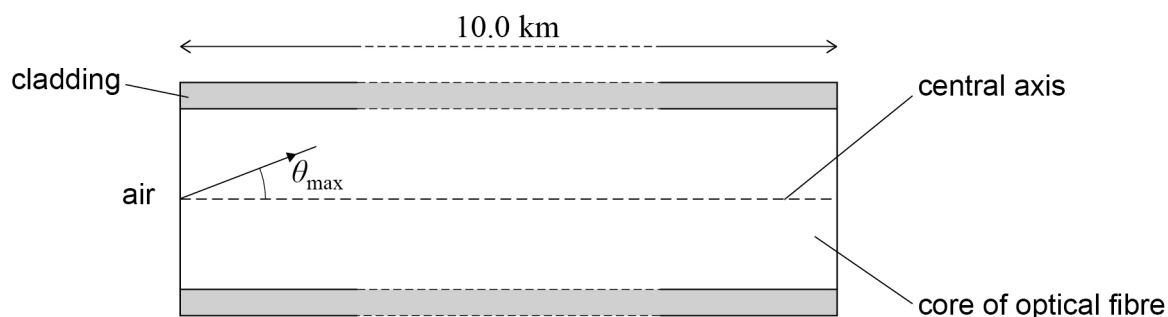
[1 mark]

refractive index = _____

A pulse of light is passed along the optical fibre and experiences modal dispersion. The fibre is straight and has a length of 10.0 km.

The longest path of light along the fibre is at an angle θ_{max} to the central axis as shown in **Figure 9**.

The shortest path of light along the fibre is along the central axis.

Figure 9**not to scale**

There is a time delay of $13.7 \mu\text{s}$ between the arrival of the pulse when it follows the longest path and when it follows the shortest path.

0 8 . 3

Show that the length of the longest path is approximately 12.7 km.

[2 marks]

0 8 . 4

Calculate θ_{max} .

[1 mark]

$\theta_{\text{max}} =$ _____ °

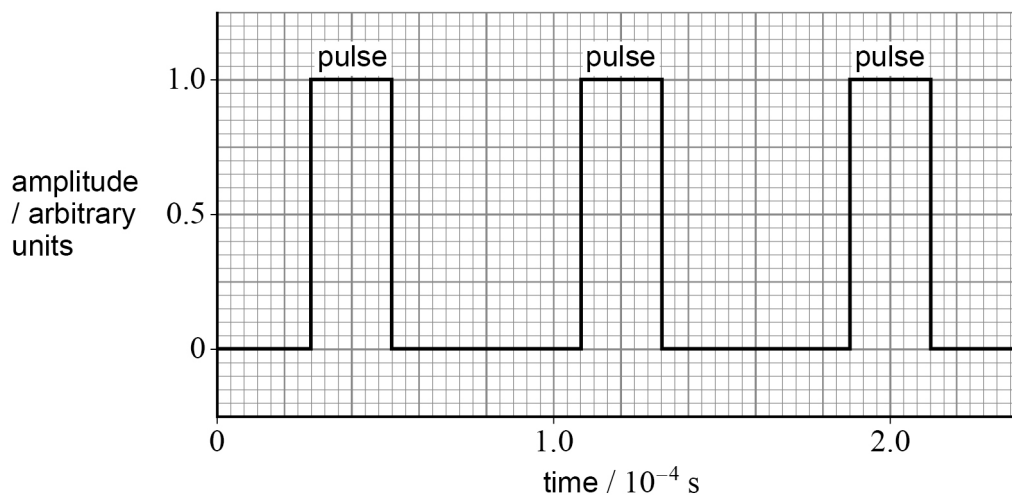
Question 8 continues on the next page



A laser sends a signal along the optical fibre. The signal is made up of a series of repeated pulses as shown in **Figure 10**.

Figure 10 shows the signal as it enters the fibre.

Figure 10



0 8 . 5

Calculate the frequency of repetition of the pulses.

[2 marks]

frequency = _____ Hz

0 8 . 6

Calculate the maximum duration of each pulse that is received at the end of this fibre.

[2 marks]

maximum duration = _____ s



0 8 . 7

Explain how pulse broadening limits the frequency at which pulses in a signal can be received through this optical fibre.

[1 mark]

11

Turn over for the next question

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09

In an experiment, a beam of electrons is accelerated from rest through a potential difference.

The beam is split into two coherent beams by a device that acts as a double slit. The two electron beams interfere to form an interference pattern as shown in **Figure 11**.

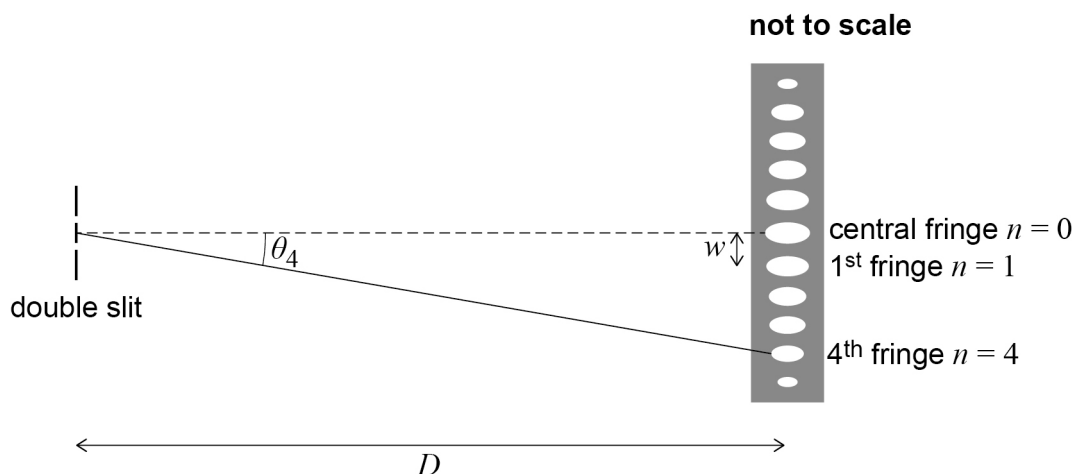
The path from the centre of the double slit to the n^{th} bright fringe makes an angle θ_n with the path to the central fringe.

The fringe separation is w .

The distance between the double slit and the interference pattern is D .

D is much larger than w .

Figure 11



09.1

Explain why θ_n in radians is approximately equal to $\frac{nw}{D}$.

[1 mark]



Figure 12 shows the interference pattern from this experiment.
The scale of the image is shown.

Figure 12



The slit separation of the double slit is 496 nm.

09.2 Show that the de Broglie wavelength of the electrons is approximately $1 \times 10^{-11} \text{ m}$.

In your answer:

- annotate **Figure 12** to show how you made your measurements
- use the relationship given in Question **09.1**.

[4 marks]

Question 9 continues on the next page

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0 9 . 3

Calculate the accelerating potential difference for the electrons.
Ignore relativistic effects.

[3 marks]

accelerating potential difference = _____ V

8

END OF SECTION A



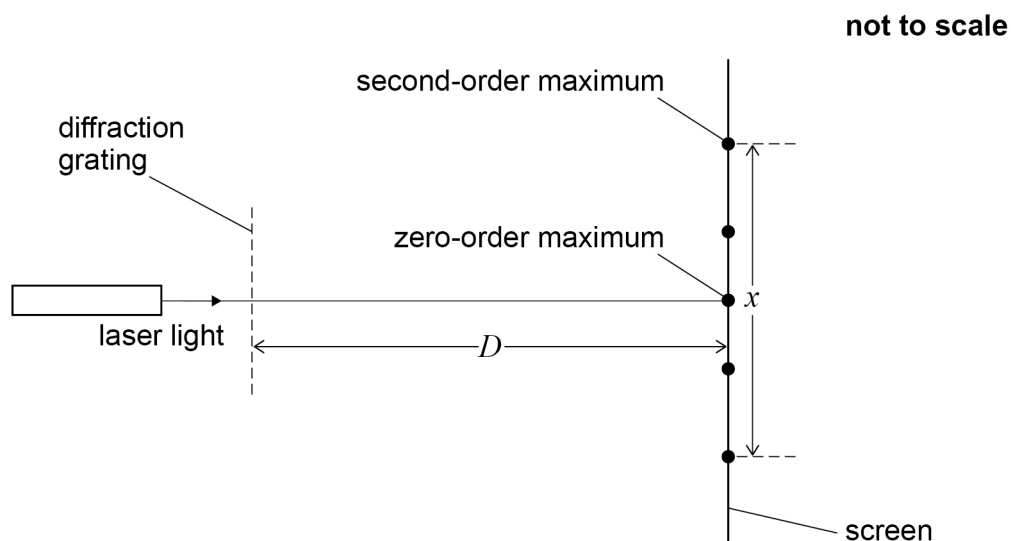
Section B

Answer **all** questions in this section.

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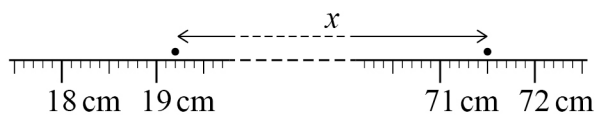
1 1

Figure 14 shows an arrangement used to determine the wavelength of laser light. The light is incident normally on a diffraction grating. The diffraction pattern is viewed on a screen a distance D from the diffraction grating.

Figure 14

The distance between the two second-order maxima is x .

A student uses a metre ruler to measure x , as shown in **Figure 15**.

Figure 15

1 1 . 1

Determine x using **Figure 15**.
State an appropriate unit for your answer.

[1 mark]

$x =$ _____

unit = _____



1 1 . 2

Calculate the percentage uncertainty in your value of x .

[1 mark]

percentage uncertainty = _____

1 1 . 3

The student measures D as 200.0 cm.
The diffraction grating has 100 lines per mm.

Calculate, in nm, the wavelength λ of the laser light.

[3 marks]

$\lambda =$ _____ nm

1 1 . 4

The diffraction grating is now replaced with a grating that has a larger number of lines per mm. The uncertainty in the grating spacing is negligible.

Discuss how this will affect the percentage uncertainty in the measured value of λ .

[2 marks]

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

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Monochromatic light is incident normally on a diffraction grating.

The first-order maximum is observed at an angle of 18° to the zero-order maximum.

What is the largest angle at which a maximum can be found?

[1 mark]

A 90°

☐

B 72°

☐

C 68°

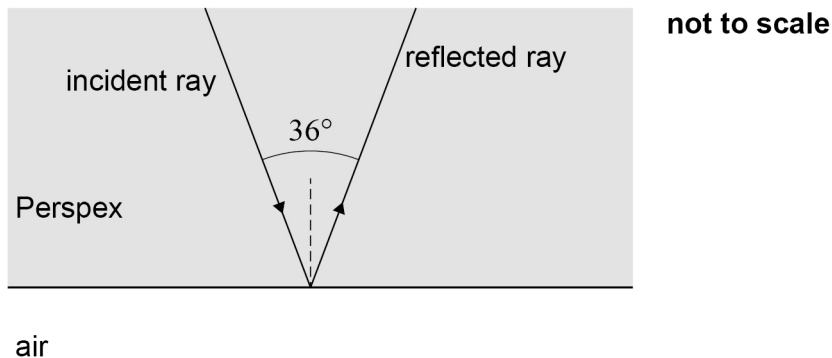
☐

D 54°

☐

2 3

Light is incident on a Perspex–air boundary and is partially reflected and partially refracted. The incident and reflected rays are shown. The refracted ray is not shown.



The reflected ray is deflected by a total angle of 36° from the incident ray.
The refractive index of Perspex is 1.5

What is the angle of refraction in air for the refracted ray?

[1 mark]

A 12°

☐

B 23°

☐

C 28°

☐

D 62°

☐

Don't write

07

Monochromatic light from a laser passes through the centre of a narrow single slit. The light forms a pattern on a screen.

07.1

Describe the appearance of the pattern.

You may support your answer with a labelled diagram.

[2 marks]

07.2

The width of the slit is decreased and a new pattern is observed on the screen.

Describe how this new pattern is different from the pattern in Question **07.1**.

[2 marks]



0 7 . 3

The single slit is replaced with a diffraction grating that has 400 lines per mm.
The wavelength of the laser light is 530 nm.
A diffraction pattern is produced.

Calculate the maximum order of this diffraction pattern.

[2 marks]

maximum order = _____

6

Turn over for the next question

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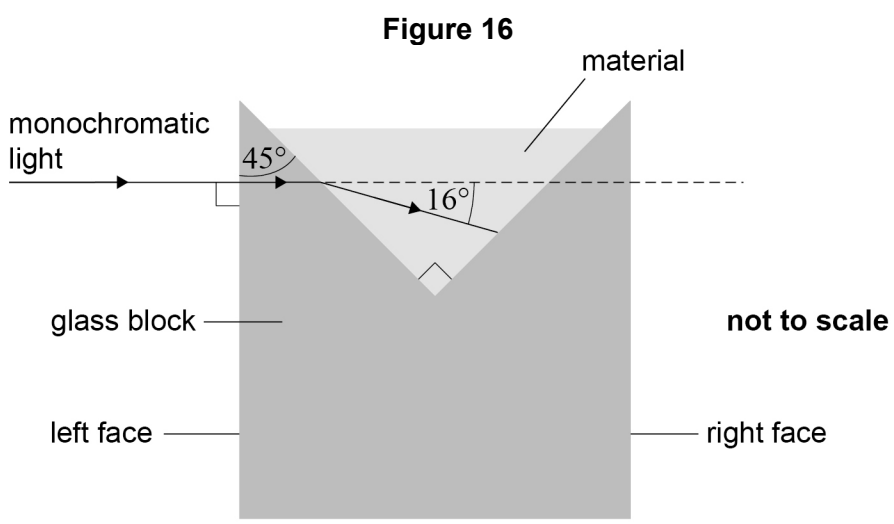
1 1

Figure 16 shows apparatus used to determine the refractive index of a material.

The material is placed between two internal faces of a glass block. The angle between the two internal faces is 90° . The angle between the vertical face and the angled face of the block is 45° .

A horizontal ray of monochromatic light travelling in air is incident at 90° to the left face of the block.

The refractive index of the glass is 1.52



1 1 . 1

Calculate the change in the speed of light as it passes from air to the glass.

[2 marks]

change in speed of light = _____ m s^{-1}

1 1 . 2

The light deviates by 16° from the horizontal as it passes from the glass to the material, as shown in **Figure 16**.

Calculate the refractive index of the material.

[3 marks]

refractive index = _____



This apparatus cannot be used to determine refractive indices smaller than a minimum value $n_{\min}$.

1 1 . 3

Determine, by considering the critical angle, the value of $n_{\min}$.

[1 mark]

$$n_{\min} = \underline{\hspace{5cm}}$$

1 1 . 4

Explain why refractive indices smaller than $n_{\min}$ cannot be measured with this apparatus.

[3 marks]

9

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

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For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.

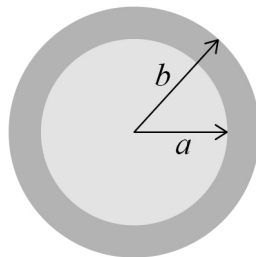


If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

A step-index optical fibre has a core of radius a and a cladding of maximum radius b .

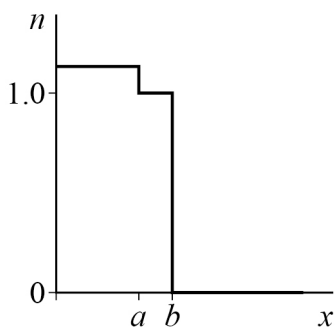


The fibre is in air.

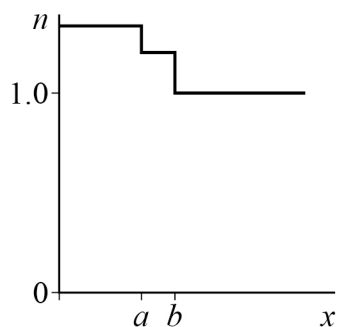
Which graph shows how the refractive index n varies with distance x from the centre of the fibre?

[1 mark]

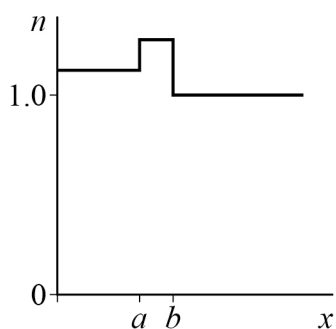
A



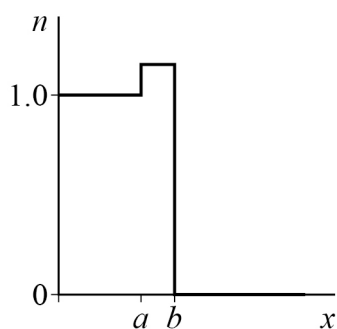
B



C



D



A



B



C



D



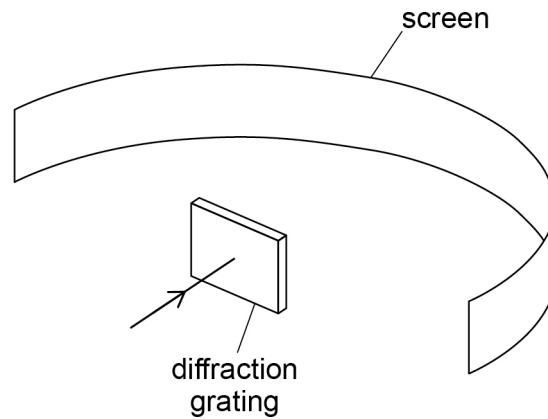
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Laser light of wavelength $6.94 \times 10^{-7} \text{ m}$ is incident normally on a diffraction grating.

The grating has 280 lines per millimetre.

Bright spots appear on a semicircular screen, as shown in **Figure 1**.

Figure 1



Calculate the total number of bright spots that appear on the screen.

[3 marks]

number = _____

Do not write

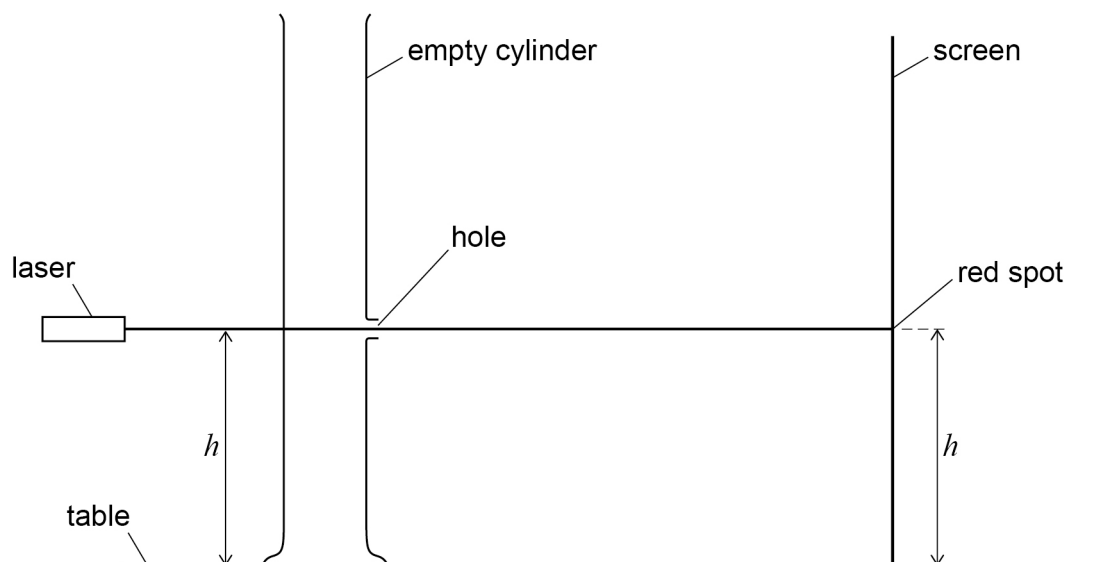
Figure 8 shows apparatus used to demonstrate refraction.

A glass measuring cylinder has a small hole on one side, at height h .

The teacher shines red laser light horizontally at height h through the wall of the empty cylinder.

The light passes through the hole and a red spot appears on a screen at height h .

Figure 8

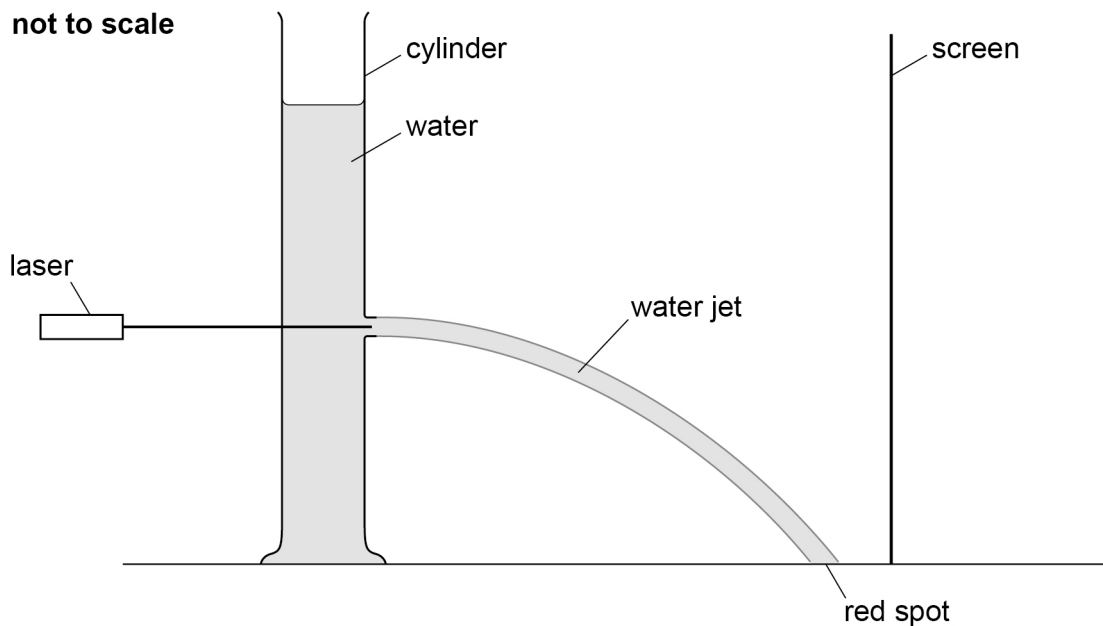


The hole is covered and the cylinder is filled with water.

The hole is then uncovered and a jet of water comes out of the hole.

The red spot disappears from the screen and appears on the table as shown in **Figure 9**.

Figure 9



0 7 . 1 Sketch, on **Figure 9**, the path of the light through the water jet.

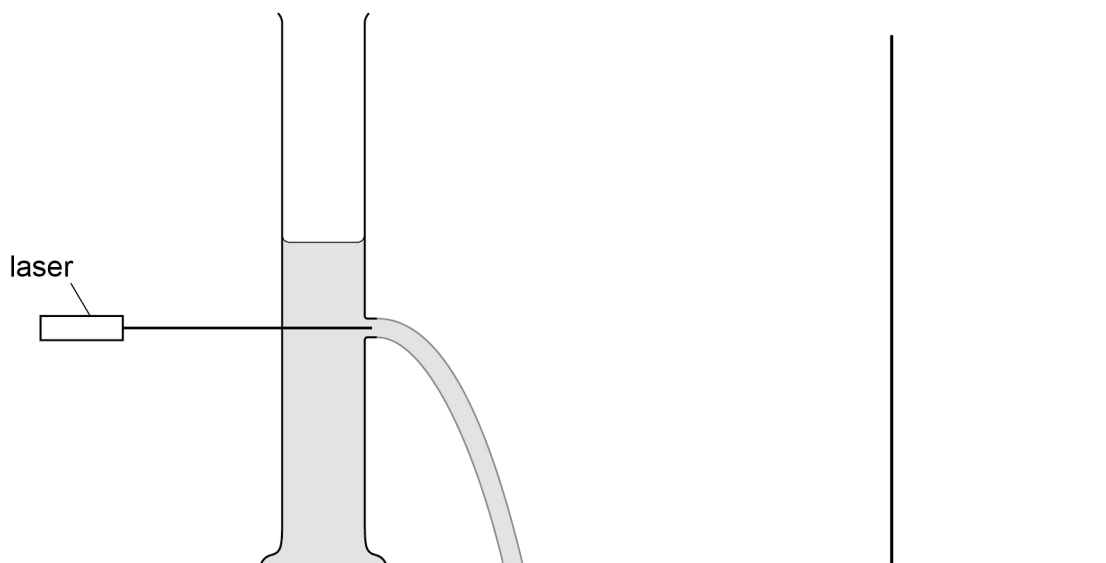
[1 mark]

Question 7 continues on the next page

Figure 10 shows the water jet when there is less water in the cylinder.

At a particular water level, the red spot disappears from the table and reappears on the screen.

Figure 10



0 7 . 2 Draw, on **Figure 10**, the new path of the light from the hole to the screen.

[1 mark]

0 7 . 3 Explain why the spot disappears from the table and reappears in the position you drew on **Figure 10**.

[3 marks]



0

7

4

State **one** safety measure when using a laser in a school laboratory.

[1 mark]

*Do not write
outside the
box*

6

2**0**Do not write
outside the
box

Light from a single monochromatic source is incident on a Young's double-slit apparatus. Fringes with spacing y are observed on a screen.

The slit separation and the distance between the slits and the screen are both doubled.

What is the new fringe spacing?

[1 mark]

A $4y$



B $2y$



C y



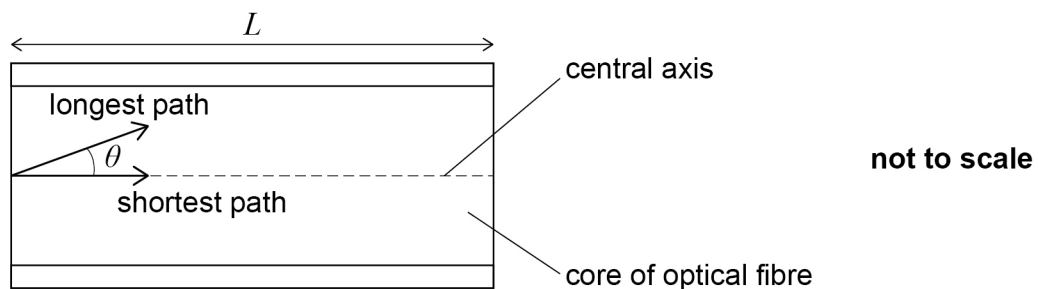
D $\frac{y}{2}$



Light travels with a speed v in the core of an optical fibre of length L .

The longest path is at an angle θ to the central axis of the fibre.

The shortest path is along the central axis of the fibre.



What is the difference between the times taken for the light to travel along the longest path and along the shortest path?

[1 mark]

A $\frac{L}{v} \left(\frac{1}{\sin \theta} - 1 \right)$ ☐

B $\frac{L}{v} (1 - \sin \theta)$ ☐

C $\frac{L}{v} \left(\frac{1}{\cos \theta} - 1 \right)$ ☐

D $\frac{L}{v} (1 - \cos \theta)$ ☐

Turn over for the next question



Do not write

The refractive indices of some materials are given below.

Material	Refractive index
Teflon	1.4
acrylic	1.5
sapphire	1.7

Materials are chosen for the core and cladding of a step-index optical fibre.

Which combination results in the least modal dispersion?

[1 mark]

	Core	Cladding	
A	Teflon	sapphire	☐
B	sapphire	Teflon	☐
C	acrylic	Teflon	☐
D	Teflon	acrylic	☐

Light of wavelength $5.5 \times 10^{-7} \text{ m}$ is incident normally on a diffraction grating.
The grating has $250 \text{ lines mm}^{-1}$.

Calculate the angle between the central maximum and a second-order maximum.

[2 marks]

angle = _____ °



Do not write

0 4

An optical fibre has a core and cladding.

The refractive index of the core is 1.49

The critical angle for the core–cladding boundary is 62.4° .

0 4 . 1

Calculate the refractive index of the cladding.

[1 mark]

refractive index = _____

0 4 . 2

An optical fibre with cladding is used to carry information using a series of pulses of monochromatic light.

This optical fibre has a larger critical angle than an optical fibre without cladding.

Explain the benefit of a larger critical angle in this optical fibre.

[2 marks]

3

Turn over for the next question

Turn over ►



Do not write

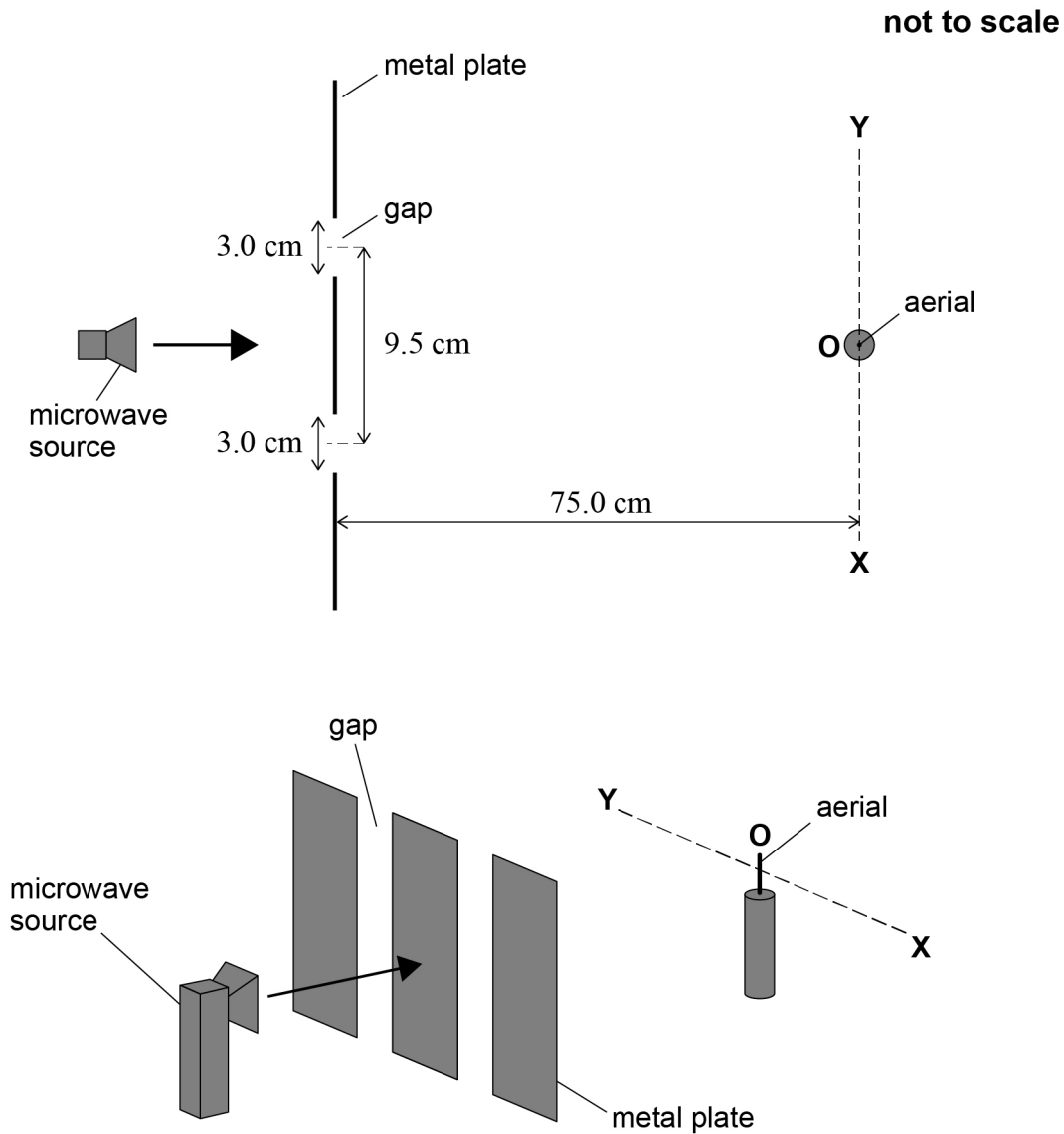
Figure 7 shows two views of apparatus used to demonstrate double-slit interference using microwaves.

The apparatus includes a microwave source, three metal plates mounted vertically, and a rod-shaped aerial. The aerial is used to measure the intensity of microwaves.

The aerial is moved parallel to the plates along the line **XY** and the microwave intensity at different positions is measured.

O is the position of the central maximum.

Figure 7



The 3.0 cm gaps between the plates act as a double slit. The distance between the centres of the slits is 9.5 cm.



The aerial is placed at **O**, a perpendicular distance of 75.0 cm from the slits.
The aerial measures a maximum intensity.

The frequency of the microwaves is 11 GHz.

A student attempts to predict the distance w along **XY** between **O** and the next maximum.

She uses the equation $w = \frac{\lambda D}{s}$ to calculate w .

0 8 . 1

Calculate w , using this equation.

[2 marks]

$w =$ _____ m

0 8 . 2

Suggest why the use of this equation does **not** give an accurate value for w in this demonstration.

[1 mark]

Question 8 continues on the next page

Turn over ►

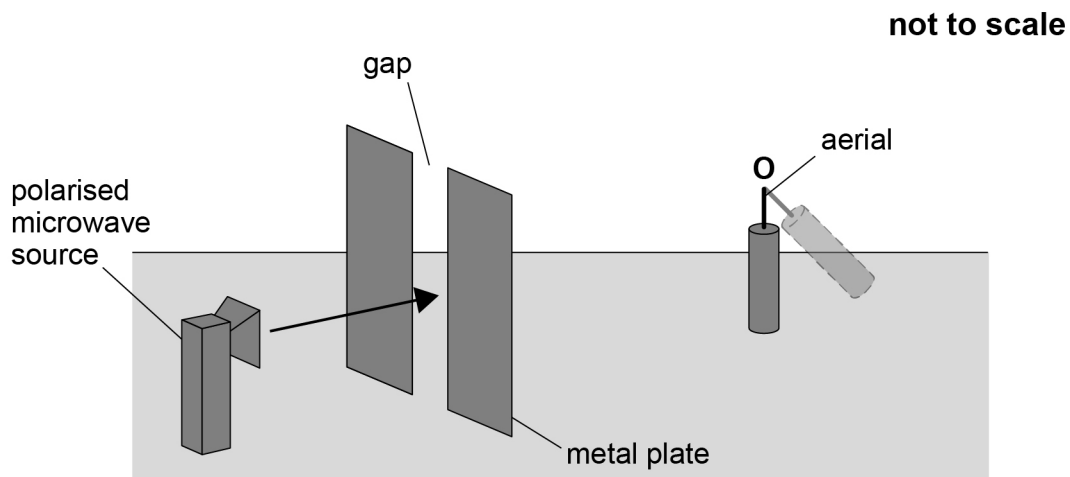


The same microwave source is used to demonstrate polarisation, as shown in **Figure 8**.

The source produces vertically polarised microwaves.

There is now just one gap between plates.

Figure 8



The aerial is rotated at **O** in a plane parallel to the plates.

The intensity is measured at different angles of rotation from 0° to 360° .



08.3

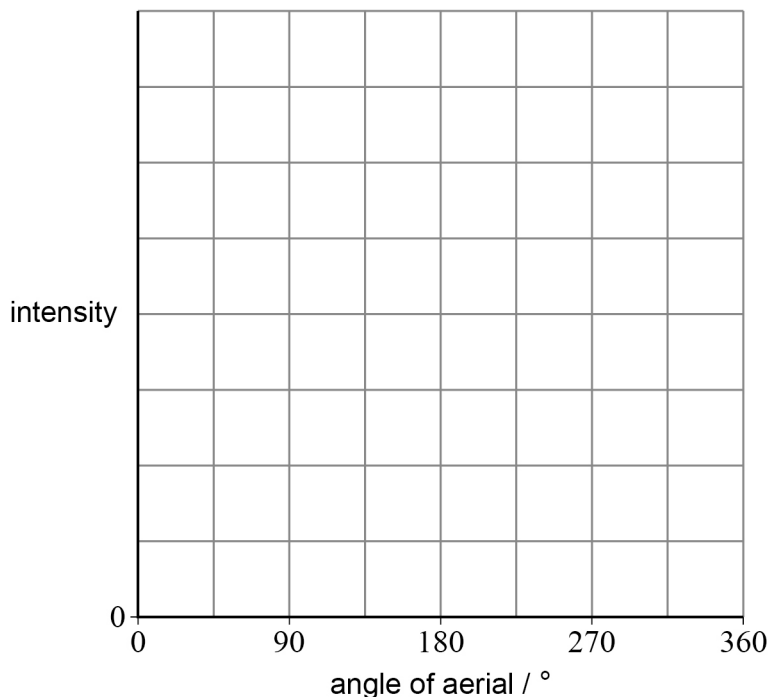
Sketch on **Figure 9** a line to show a relationship between the measured intensity and the angle of the aerial.

Label this line **A**.

Values are not required on the y -axis.

[1 mark]

Figure 9



08.4

Explain your answer to Question **08.3**.

[2 marks]

08.5

The microwave source is replaced with a new microwave source that produces unpolarised microwaves. These microwaves have the same intensity as those produced in **Figure 8**.
The experiment is then repeated.

Sketch on **Figure 9** a line to show the relationship between the measured intensity and the angle of the aerial for this experiment.

Label this line **B**.

[2 marks]



Do not write

1**9**

A student is using a laser to investigate diffraction in a school laboratory.

Which procedure would **not** be suitable for this experiment?

[1 mark]

A using a laser with a wavelength in the visible part of the spectrum

☐

B directing the lasers so that patterns are produced at eye level

☐

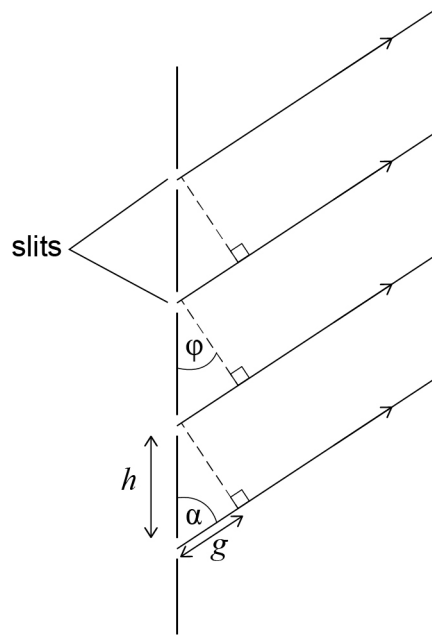
C taking a photograph of the fringes next to a ruler

☐

D projecting the laser light onto a wall

☐

2 1 In the derivation of the diffraction grating equation $n \lambda = d \sin \theta$, a diagram is drawn similar to the one below. The arrows show the rays that meet to form the first-order maximum.



Which row shows the angle that is equal to θ and the distance that is equal to λ ?

[1 mark]

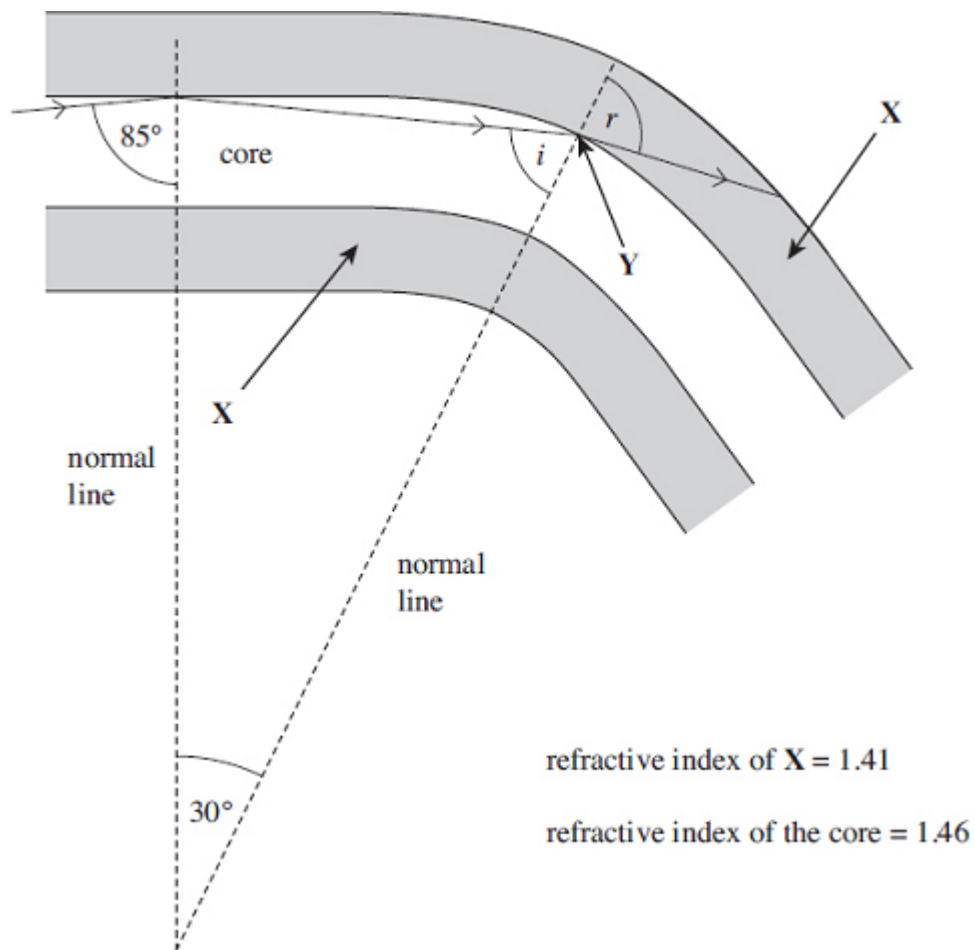
	Angle equal to θ	Distance equal to λ	
A	ϕ	g	☐
B	α	h	☐
C	α	g	☐
D	ϕ	h	☐



Do not write

Figure 1 shows a cross-section through an optical fibre used for communications.

Figure 1



. 1 Identify the part of the fibre labelled X.

[1 mark]

0 1 . 2 Calculate the critical angle for the boundary between the core and X.

[2 marks]

angle = _____ degrees

0 1 . 3 The ray leaves the core at Y. At this point the fibre has been bent through an angle of 30° as shown in **Figure 1**.

Calculate the value of angle i .

[1 mark]

angle = _____ degrees

0 2 . 1 A metal wire of length 1.4 m has a uniform cross-sectional area of $7.8 \times 10^{-7} \text{ m}^2$.

Calculate the resistance, R , of the wire.
Resistivity of the metal = $1.7 \times 10^{-8} \Omega \text{ m}$

[2 marks]

resistance = _____ Ω

0 2 . 2 The wire is now stretched to twice its original length by a process that keeps its volume constant. The resistivity of the metal of the wire remains constant.

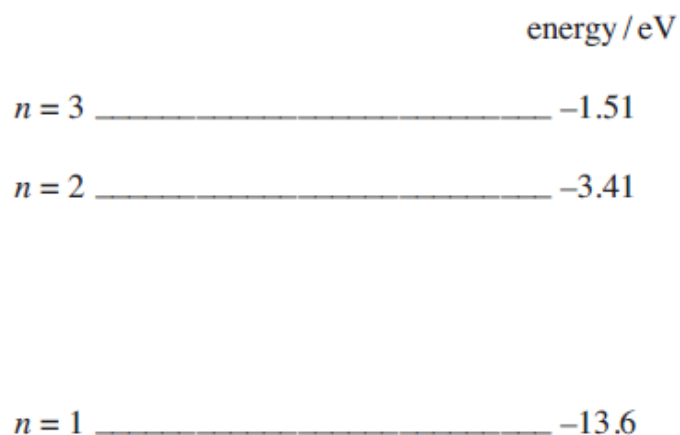
Show that the resistance increases to $4R$.

[2 marks]

0	3
---	---

Figure 2 shows the lowest three energy levels of a hydrogen atom.

Figure 2



An electron is incident on a hydrogen atom. As a result an electron in the ground state of the hydrogen atom is excited to the $n = 2$ energy level. The atom then emits a photon of a characteristic frequency.

Explain how the electron in the ground state becomes excited to the $n = 2$ energy level.

[2 marks]

0	4
---	---

A battery in a laptop computer has an electromotive force (emf) of 14.8 V and can store a maximum charge of 15.5×10^3 C. The battery has negligible internal resistance.

Calculate the maximum amount of energy this battery can deliver.

[2 marks]

maximum energy = _____ J

0	5
---	---

When a mass M attached to a spring X , as shown in **Figure 3**, is displaced downwards and released it oscillates with time period 1.5 s .

In **Figure 4**, a second identical spring is connected as shown and the same mass M is attached.

Figure 3

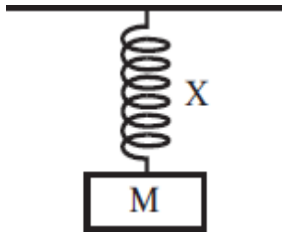
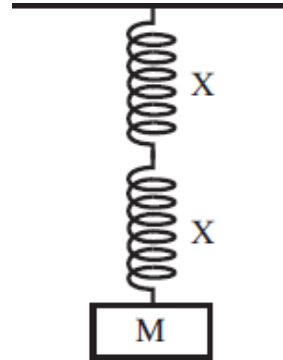


Figure 4



Calculate the time period of the mass in **Figure 4** when it is displaced downwards and released.

[2 marks]

time period = _____ s

0	6
---	---

Figure 5 represents a stationary wave formed on a steel string fixed at **P** and **Q** when it is plucked at its centre.

Figure 5



The stationary wave in **Figure 5** has a frequency of 150 Hz . The string **PQ** has a length of 1.2 m .

Calculate the wave speed of the waves forming the stationary wave.

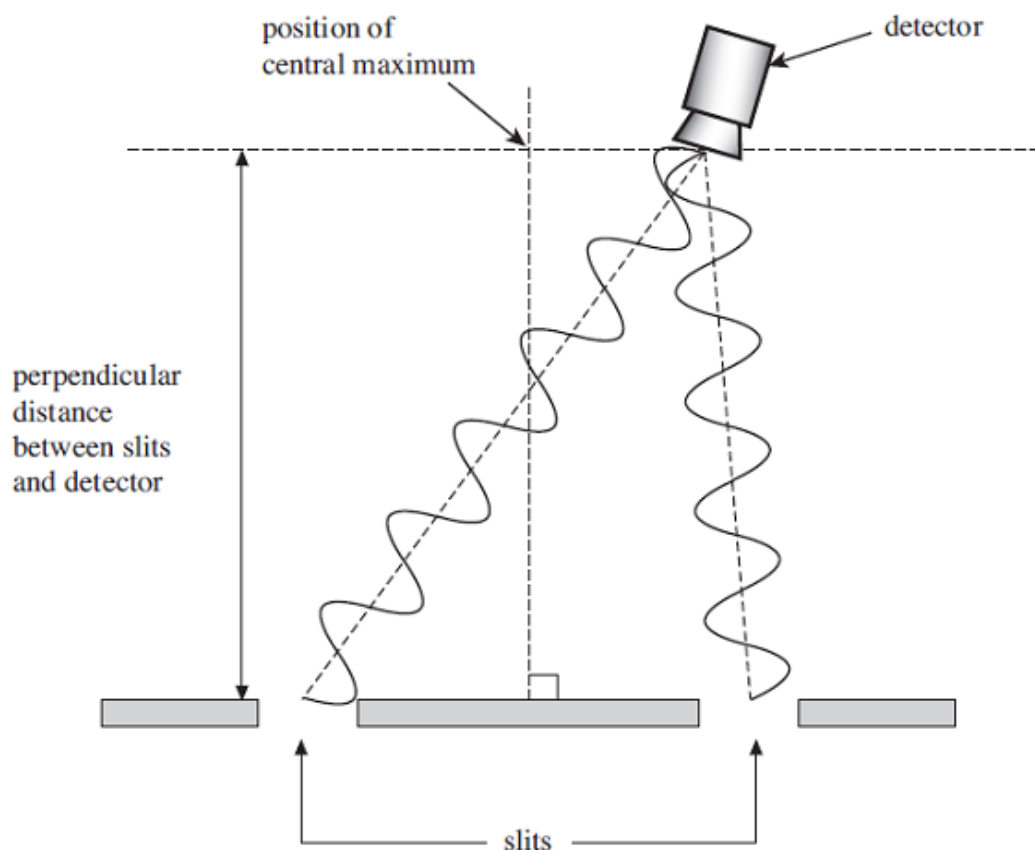
[2 marks]

wave speed = _____ m s^{-1}

07

Figure 6 shows the paths of microwaves from two narrow slits through a vacuum to a detector. The slits act as coherent sources.

Figure 6



07

1

Explain what is meant by **coherent sources**.

[2 marks]

07

2

The frequency of the microwaves is 9.4 GHz.

Determine the wavelength of the waves.

[2 marks]

wavelength = _____ m

0 7 . 3 Calculate the path difference between the two waves arriving at the detector.
Use the information on **Figure 6** and your answer to **7.2**.
[1 mark]

path difference = _____ m

0 7 . 4 Explain whether a maximum or minimum is detected at the position shown in **Figure 6**.
[3 marks]

0 7 . 5 The experiment is now rearranged so that the perpendicular distance from the slits to the detector is 0.42 m. The interference fringe spacing changes to 0.11 m.
Calculate the slit separation.
Give your answer to an appropriate number of significant figures.
[3 marks]

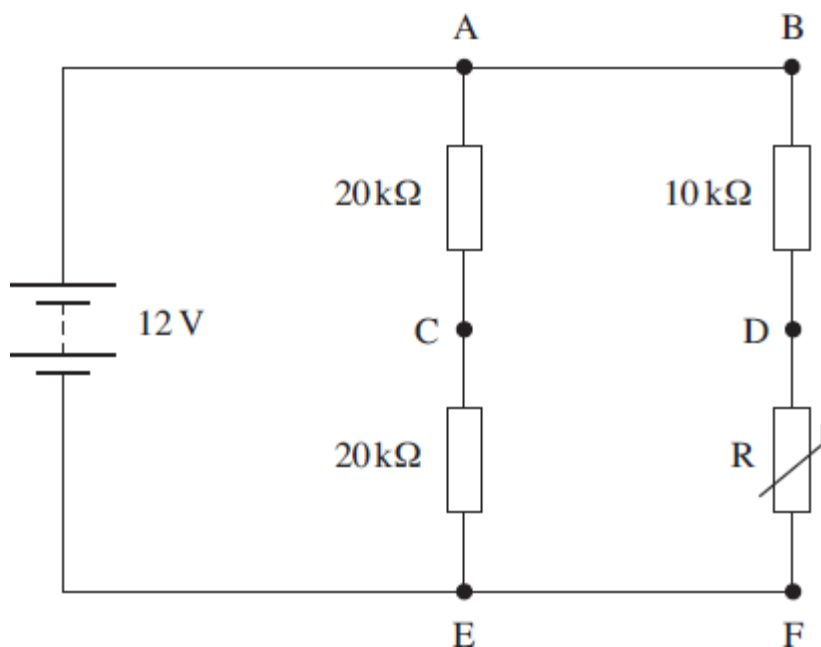
slit separation = _____ m

0 7 . 6 With the detector at the position of a maximum, the frequency of the microwaves is now doubled.
Explain what would now be detected by the detector in the same position.
[2 marks]

0 8

Figure 7 shows a 12 V battery of negligible internal resistance connected to a combination of three resistors and a thermistor.

Figure 7



0 8

. 1

At room temperature, the resistance of the thermistor is $5.0\text{ k}\Omega$.

Calculate the total resistance of the circuit.

[2 marks]

total resistance = _____ $\text{k}\Omega$

0 8

. 2

Calculate the current in the battery at the same temperature.

[1 mark]

current = _____ A

0 8 . 3

A high-resistance voltmeter is used to measure the potential difference (pd) between points **A–C**, **D–F** and **C–D** in turn.

Complete **Table 1** indicating the reading of the voltmeter at each of the three positions.

[3 marks]

Table 1

Voltmeter position	pd / V
A–C	
D–F	
C–D	

0 8 . 4

The thermistor is heated so that its resistance decreases.

State and explain the effect this has on the voltmeter reading in the following positions.

[4 marks]

A–C _____

D–F _____

Turn over for the next question

09

Experiments based on the photoelectric effect support the particle nature of light. In such experiments, light is directed at a metal surface. A threshold frequency for this light occurs. Below this frequency, no photoelectric effect is observed.

09 . 1

Explain why the photoelectric effect is **not** observed below the threshold frequency.

[2 marks]

09 . 2

Monochromatic light of wavelength $5.40 \times 10^{-7} \text{ m}$ is incident on a metal surface which has a work function of $1.40 \times 10^{-19} \text{ J}$.

Calculate the energy of a single photon of this light.

[2 marks]

energy = _____ J

09 . 3

Calculate the maximum kinetic energy of an electron emitted from the surface.

[1 mark]

energy = _____ J

09 . 4

Calculate the maximum speed of the emitted electron.

[2 marks]

maximum speed = _____ m s^{-1}

0 9 . 5

Calculate the de Broglie wavelength of the fastest electrons.

[2 marks]

wavelength = _____m

0 9 . 6

The intensity of light incident on the metal is increased without changing the frequency.

Describe and explain the change which takes place.

[2 marks]

Turn over for the next question

Section B

Answer **all** questions in this section.

1 0

A student carries out an experiment to determine the emf ε and internal resistance r of a cell. The cell is connected to a variable resistor R . The current through the cell and the terminal pd of the cell are measured as the resistance of R is decreased. The voltmeter used had a very high resistance. The circuit is shown in **Figure 8**.

Figure 8

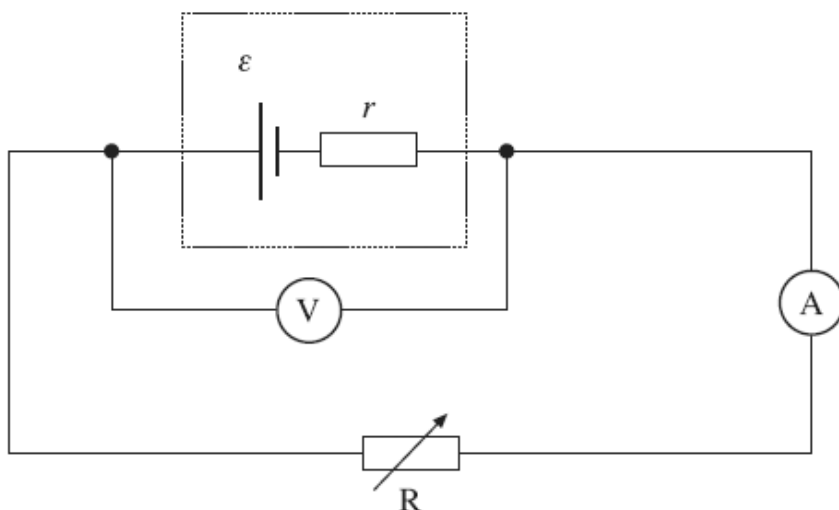
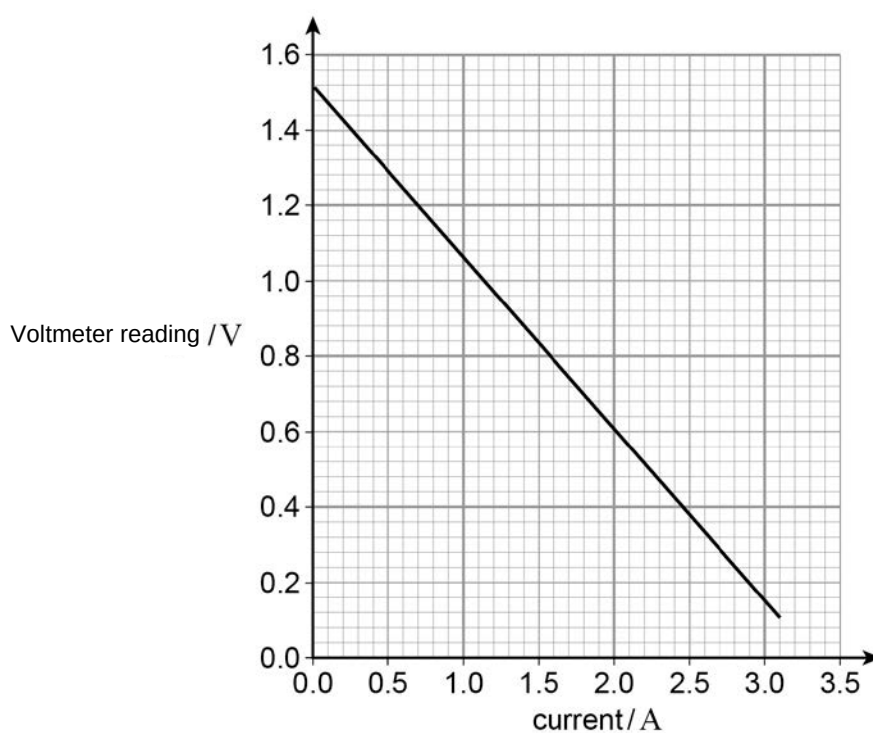


Figure 9 shows a graph drawn from the results from the experiment.

Figure 9



10 . 1

Determine the emf $\mathcal{E}$ of the cell using **Figure 9**.

[1 mark]

emf of cell = _____ V

10 . 2

Determine the internal resistance r of the cell by using **Figure 9**.

[3 marks]

internal resistance = _____ Ω

10 . 3

Another student checks the emf of the cell by disconnecting the cell from the circuit and connecting a voltmeter across it. The voltmeter used has a low resistance.

State and explain how this value compares with the value determined in **10.1**.

[2 marks]

Turn over for the next question

1 1

Read through the following information and answer the questions that follow.

Measuring the speed of sound in air

1

Sound travels as a longitudinal wave. The first attempt to measure its speed in air was made by the scientist Gassendi. The procedure involved timing the interval between seeing the flash of a gun and hearing the bang from some distance away. Gassendi assumed that the speed of light is infinite compared to the speed of sound. The value he obtained for the speed of sound was 480 m s^{-1} . He also realised that the speed of sound does not depend on frequency.

5

1 1 . 1

Suggest an experiment that will demonstrate the wave nature of sound (line 1).

[1 mark]

1 1 . 2

Using Gassendi's value for the speed of sound (line 5), calculate the time between seeing the flash of a gun and hearing its bang over a distance of 2.5 km.

[1 mark]

time = _____ s

1 1 . 3

Explain why it was necessary to assume that 'the speed of light is infinite' (line 4).

[1 mark]

1	1	.	4
---	---	---	---

Explain **one** observation that could have led Gassendi to conclude that 'the speed of sound does not depend on frequency' (line 6).

[2 marks]

Question 11 continues on the next page

Read through the following information and answer the questions that follow.

- 1 Using the same method as Gassendi, a much better value of 350 m s^{-1} was obtained by the Italian physicists Borelli and Viviani. In 1740 another Italian, Bianconi, showed that sound travels faster when the temperature of the air is greater. In 1738 a value of 332 m s^{-1} was obtained by scientists in Paris. This is very close to the currently accepted value considering the measuring equipment available to the scientists at that time. Since 1986 the accepted value has been 331.29 m s^{-1} at 0°C .

- 1 1 . 5 Explain how the value obtained by Borelli and Viviani was 'much better' than that obtained by Gassendi (line 1).

[1 mark]

- 1 1 . 6 The speed of sound c in dry air is given by

$$c = k\sqrt{(\theta + 273.15)}$$

where θ is the temperature in $^\circ\text{C}$, and k is a constant.

Calculate a value for k using data from the passage.

[2 marks]

$$k = \text{_____} \text{ m s}^{-1} \text{ K}^{-1/2}$$

- 1 1 . 7 State the steps taken by the scientific community for the value of a quantity to be 'accepted' (line 5).

[2 marks]

Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C**, and **D**. For each question select the best response.

Only **one** answer per question is allowed.

For each answer completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



1 2

Two points on a progressive wave are one-eighth of a wavelength apart. The distance between them is 0.5 m, and the frequency of the oscillation is 10 Hz. What is the minimum speed of the wave?

[1 mark]

- | | | |
|----------|-----------------------|-----------------------|
| A | 0.2 m s ⁻¹ | ☐ |
| B | 10 m s ⁻¹ | ☐ |
| C | 20 m s ⁻¹ | ☐ |
| D | 40 m s ⁻¹ | ☐ |

1 3

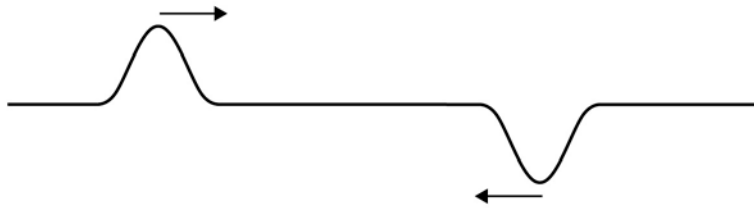
Which of the following waves **cannot** be polarised?

[1 mark]

- | | | |
|----------|-------------|-----------------------|
| A | microwave | ☐ |
| B | radio | ☐ |
| C | ultrasonic | ☐ |
| D | ultraviolet | ☐ |

1 4

The diagram shows two pulses on a string travelling towards each other.



Which of the following diagrams shows the shape of the string when the pulses have passed through each other?

[1 mark]



1 5

Monochromatic light of wavelength 590 nm is incident normally on a plane diffraction grating having $4 \times 10^5 \text{ lines m}^{-1}$. An interference pattern is produced. What is the highest order visible in this interference pattern?

[1 mark]

- A** 2 ☐
- B** 3 ☐
- C** 4 ☐
- D** 5 ☐

1 6

Sound waves cross a boundary between two media X and Y. The frequency of the waves in X is 400 Hz . The speed of the waves in X is 330 m s^{-1} and the speed of the waves in Y is 1320 m s^{-1} . What are the correct frequency and wavelength in Y?

[1 mark]

	Frequency / Hz	Wavelength / m	
A	100	0.825	☐
B	400	0.825	☐
C	400	3.30	☐
D	1600	3.30	☐

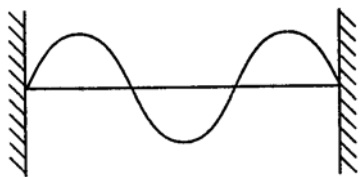
1 7

Which of the following is correct for a stationary wave?

[1 mark]

- A** Between two nodes the amplitude of the wave is constant. ☐
- B** The two waves producing the stationary wave must always be 180° out of phase. ☐
- C** The separation of the nodes for the second harmonic is double the separation of nodes for the first harmonic. ☐
- D** Between two nodes all parts of the wave vibrate in phase. ☐

1	8
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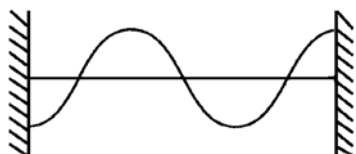


frequency of vibration = 50 Hz

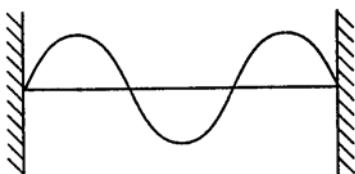
The diagram above shows a stationary wave on a stretched string at a time $t = 0$. Which one of the diagrams, A to D, correctly shows the position of the string a time $t = 0.010$ s?

[1 mark]

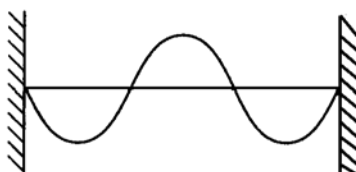
A



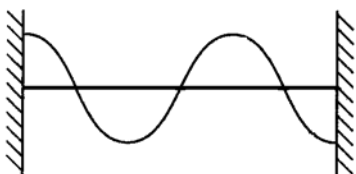
B



C



D



1 9

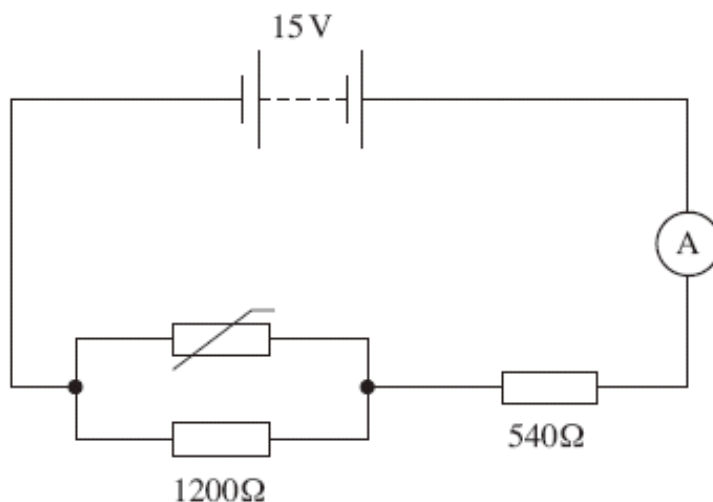
In a cathode ray tube 7.5×10^{15} electrons strike the screen in 40 s.
What current does this represent?
Charge of the electron is 1.6×10^{-19} C.

[1 mark]

- A** 1.3×10^{-16} A ☐
- B** 5.3×10^{-15} A ☐
- C** 3.0×10^{-5} A ☐
- D** 1.2×10^{-3} A ☐

2 0

The circuit shown below shows a thermistor connected in a circuit with two resistors, an ammeter and a battery of emf 15 V and negligible internal resistance.



When the thermistor is at a certain temperature the current through the ammeter is 10.0 mA.

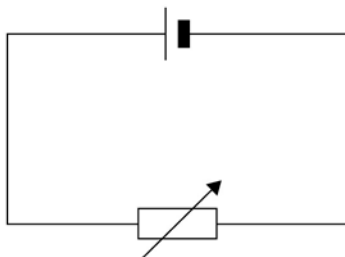
Calculate the pd across the 1200 Ω resistor.

[1 mark]

- A** 4.8 V ☐
- B** 5.4 V ☐
- C** 7.5 V ☐
- D** 9.6 V ☐

2 1

The cell in the circuit has an emf of 2.0 V. When the variable resistor has a resistance of $4.0\ \Omega$, the potential difference (pd) across the terminals of the cell is 1.0 V.



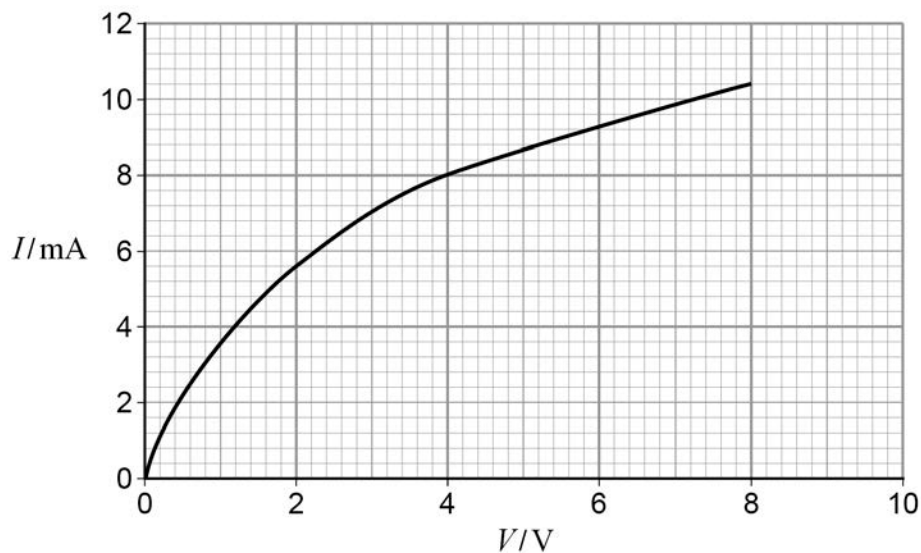
What is the pd across the terminals of the cell when the resistance of the variable resistor is $12\ \Omega$?

[1 mark]

- A 0.25 V ☐
- B 0.75 V ☐
- C 1.33 V ☐
- D 1.50 V ☐

2 2

The graph shows the current–voltage (I – V) characteristics of a filament lamp.



What is the resistance of the filament when the potential difference (pd) across it is 4.0 V?

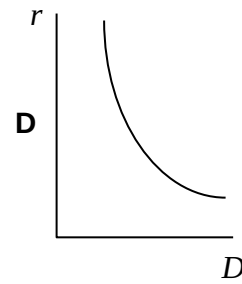
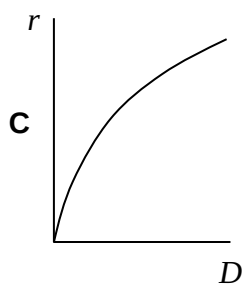
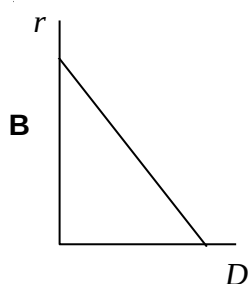
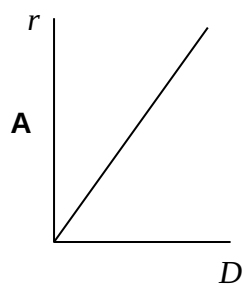
[1 mark]

- A 500 Ω ☐
- B 1700 Ω ☐
- C 2000 Ω ☐
- D 6000 Ω ☐

2 3

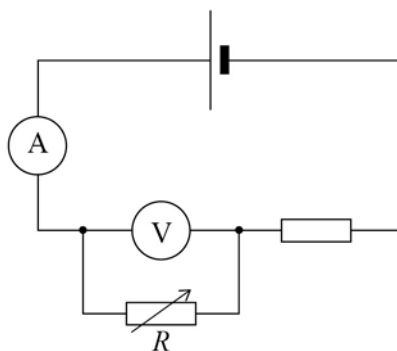
Which graph shows how the resistance per unit length r of a wire varies with diameter D of the wire?

[1 mark]

A ☐B ☐C ☐D ☐

2 4

In the circuit shown in the diagram, the cell has negligible internal resistance.



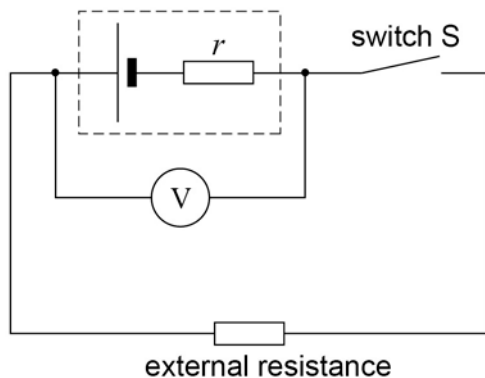
What happens to the reading of both meters when the resistance of R is decreased?

[1 mark]

	Reading of ammeter	Reading of voltmeter	
A	increases	increases	☐
B	increases	decreases	☐
C	decreases	increases	☐
D	unchanged	decreases	☐

2	5
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In the circuit shown, V is a voltmeter with a very high resistance. The internal resistance of the cell, r , is equal to the external resistance in the circuit.



Which of the following is **not** equal to the emf of the cell?

[1 mark]

- A the reading of the voltmeter when the switch S is open
- B the chemical energy changed to electrical energy when unit charge passes through the cell
- C twice the reading of the voltmeter when the switch S is closed
- D the electrical energy produced when unit current passes through the cell

☐
☐
☐
☐